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Enterprise Network Infrastructure Documentation

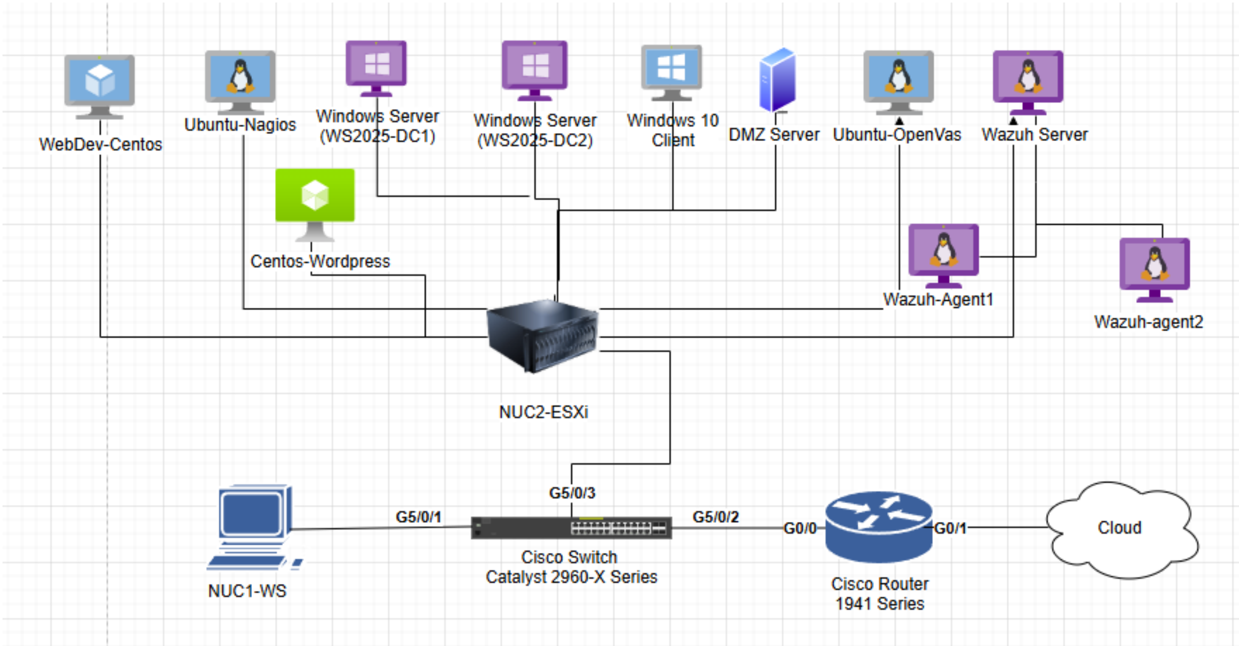
Project: Design and Implementation of a Scalable Enterprise Virtualized Network

Executive Summary

This report documents the end-to-end design and implementation of a secure, virtualized network infrastructure centered on deployment of **VMware ESXi v8** on physical NUC hardware and **Windows Server 2025** environment to provide centralized **DHCP service**. The initial phase focused on hardware provisioning and base system installation, followed by a rigorous network segmentation strategy that utilized **VLANs** and **Router-on-a-Stick (ROAS)** configurations to isolate classroom, client, server, and DMZ traffic. Following the establishment of the physical and logical network layers, the infrastructure was hardened through a **Cisco Router 1941 Series** and **Switch Catalyst 2960-X Series** configurations**,** including **SSH access** and port security, Access Control Lists **(ACLs)**, and dynamic routing through **OSPF**. A **DMZ** environment was also implemented to securely host a CentOS-based web server protected with HTTPS using a self-signed certificate.

The environment was further expanded through the deployment of Active Directory Domain Services **(AD DS)**, including a multi-domain controller configuration for centralized authentication, policy enforcement, **DNS** management, and automated user provisioning using PowerShell and CSV-based account creation. Additional enterprise authentication controls were implemented using **RADIUS** through Microsoft Network Policy Server (**NPS**). Finally, the infrastructure was extended with additional enterprise services, including **Docker, Nagios, OpenVAS, WordPress, and Wazuh** deployments.

Network Diagram



Part 1 – ESXi Deployment and Core Infrastructure Setup

1.1 Overview

The objective of part 1 was to deploy a functional virtualized infrastructure using **VMware ESXi v8** and prepare the environment for future network segmentation and services. This included the physical setup of hardware components, network configuration, installation of the ESXi hypervisor, and deployment of **Windows Server 2025** virtual machine configured as a DHCP server.

1.2 Physical Configuration

The physical infrastructure was first prepared by installing the Intel NUC hardware into the assigned rack. The Cisco switch and router within the rack were accessed via console and cleared of all prior configurations to ensure a clear state deployment. All hardware devices were physically labeled to ensure identification during troubleshooting and future expansion. Since the centralized DHCP was not yet operational, static IP was manually assigned on Classroom PC to allow communication with ESXi management interface.

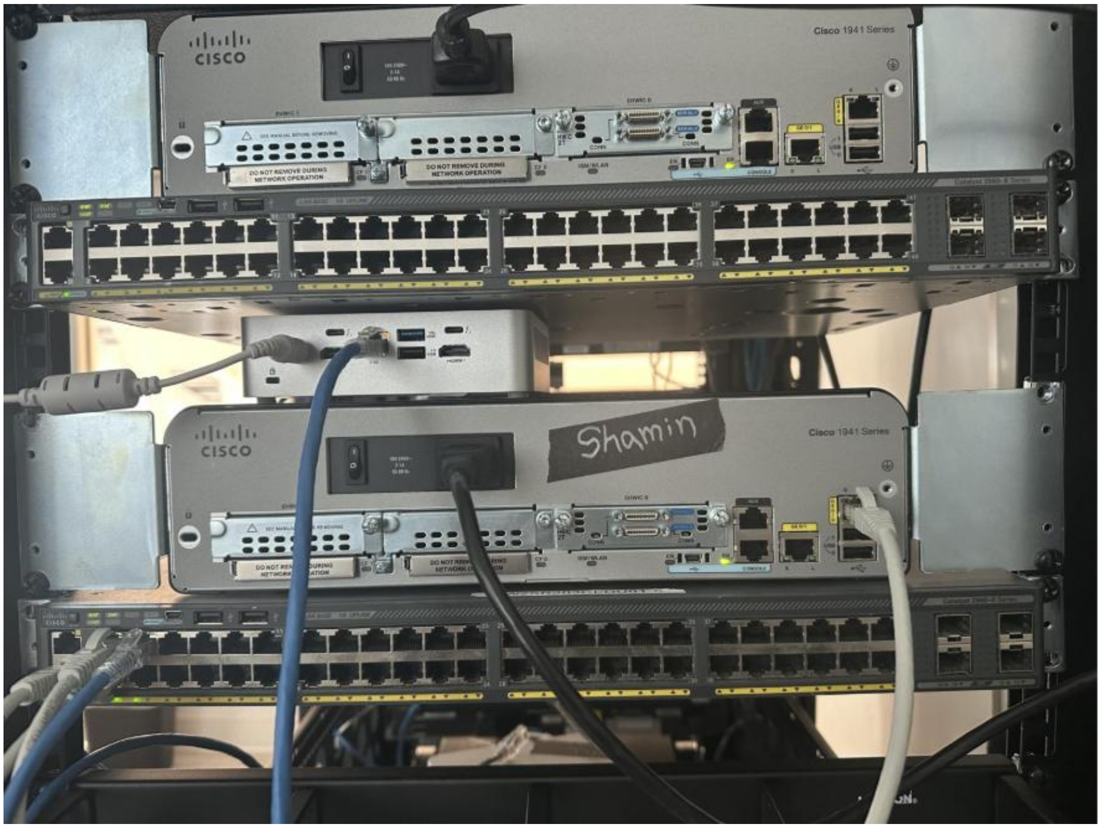


Figure 1 – Rack setup with NUC, Switch, and Router

1.3 Network Addressing Plan

Device	Type	VLAN	IP address	Subnet Mask	Default Gateway
NUC1-WS	Physical	10	10.115.10.20 (DHCP later)	255.255.255.0	10.115.10.1
Cisco Router 1941 Series	Physical	10/20/30/40/50	10.115.10.1 / 10.115.20.1 / 10.115.30.1 / 10.115.40.1 / 10.115.50.1	255.255.255.0	
Switch Catalyst 2960-X Series	Physical	40	10.115.40.2	255.255.255.0	10.115.40.1
NUC2-ESXi	Physical	40	10.115.40.10	255.255.255.0	10.115.40.1
WS2025-DC1	VM	30	10.115.30.10	255.255.255.0	10.113.30.1

WS2025-DC2	VM	30	10.115.30.11	255.255.255.0	10.115.30.1
Win10- Client	VM	20	10.115.20.10 (DHCP later)	255.255.255.0	10.115.20.1
TS-SWITCH-LINK	Physical		172.16.1.115 (router port)	255.255.255.0	
DMZ Server	VM	99	10.115.99.99	255.255.255.0	10.115.99.1
WebDev-Centos	VM	30	10.115.60.15	255.255.255.0	10.115.30.1
Centos-Wordpress	VM	30	10.115.70.20	255.255.255.0	10.115.30.1
Ubuntu-Nagios	VM	30	10.115.75.25	255.255.255.0	10.115.30.1
Ubuntu-OpenVas	VM	30	10.115.80.30	255.255.255.0	10.115.30.1
Wazuh Server	VM	30	10.115.90.35	255.255.255.0	10.115.30.1
Wazuh-agent1	VM	30	10.115.90.40	255.255.255.0	10.115.30.1
Wazuh-agent2	VM	30	10.115.90.45	255.255.255.0	10.115.30.1

1.4 ESXi Installation and Configuration

VMware ESXi was installed on the NUC via bootable USB drive to serve as the hypervisor for hosting virtual machines. During the installation, the management network was manually configured on ESXi, assigning a static IP address within physical device VLAN range.

After the installation, the successful connectivity was verified by a successful ping from Classroom PC **(NUC1-WS)** to ESXi host's management IP. Following this, the ESXi host web interface Was successfully accessed.

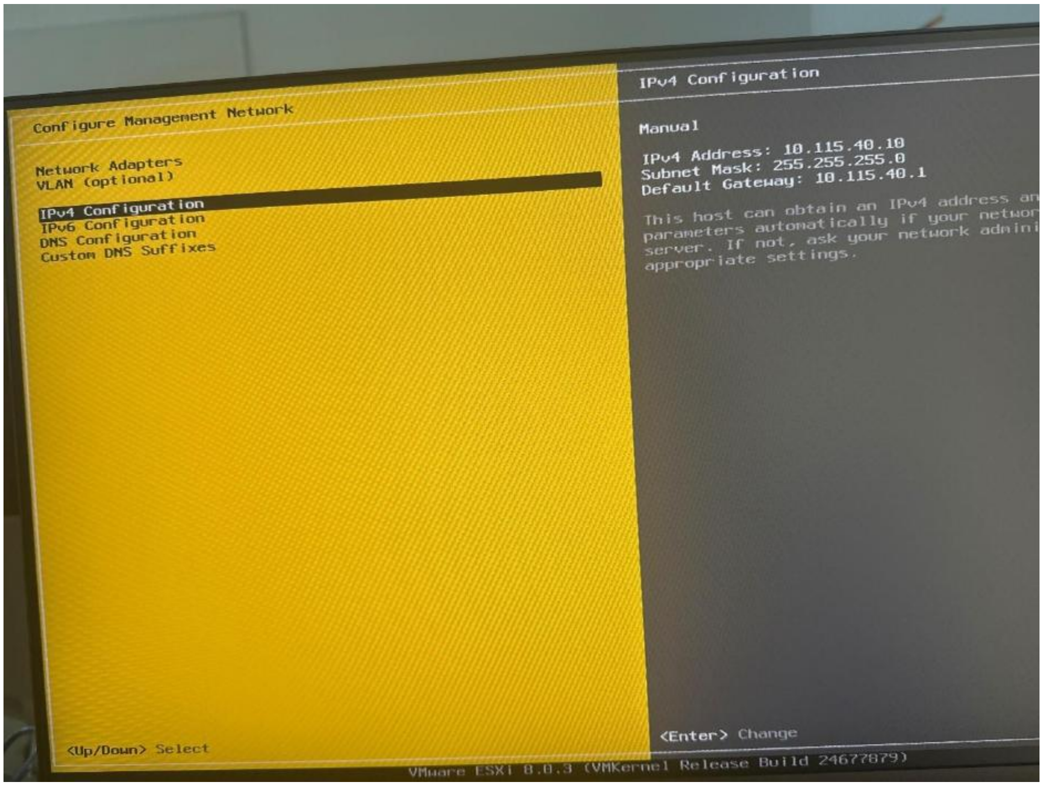


Figure 2 – Assigned static IP on ESXi

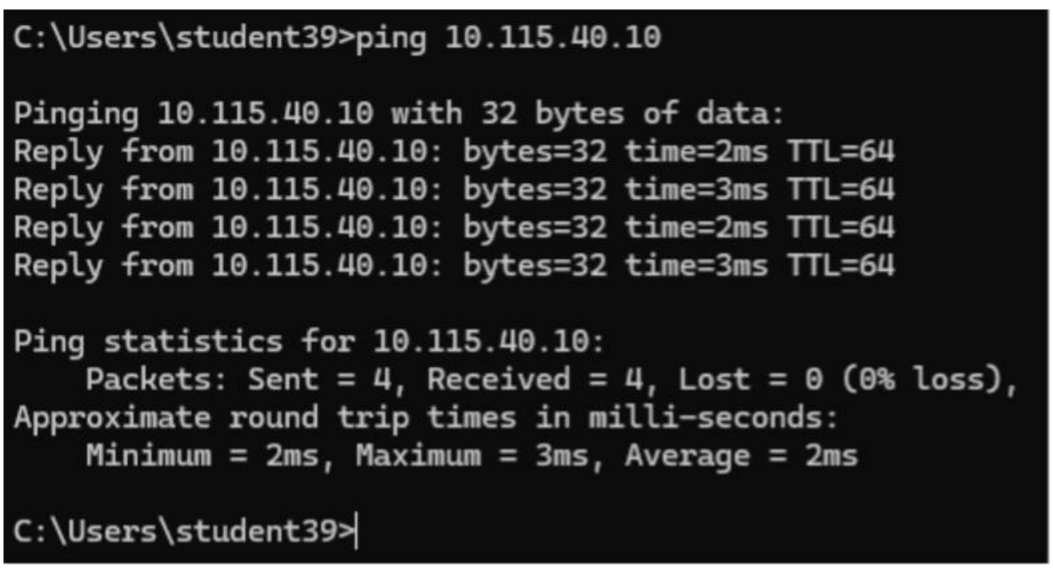


Figure 3 – Successful ping from classroom PC to ESXi host

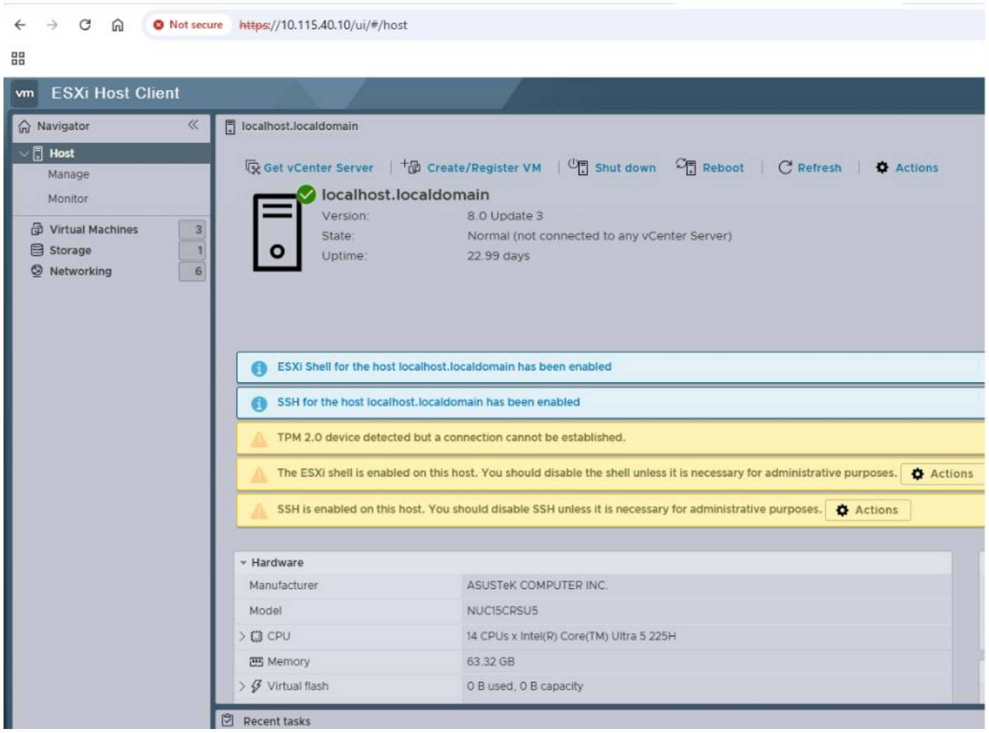


Figure 4 – ESXi host client dashboard

1.5 Windows Server 2025 Deployment and DHCP setup

A Windows Server 2025 **(WS2025-DC1)** virtual machine was installed within the ESXi environment using the provided ISO image. After installation, the system name was changed from the default to a meaningful hostname to align with standard enterprise naming conventions.

The Windows server was assigned a static IP address within the Server VLAN range **(VLAN 30).** This ensures that the server remains at a fixed point of contact for all clients. The Windows Server 2025 **(WS2025-DC1)** system was configured with the DHCP server role to enable dynamic IP addresses for client devices within the network.

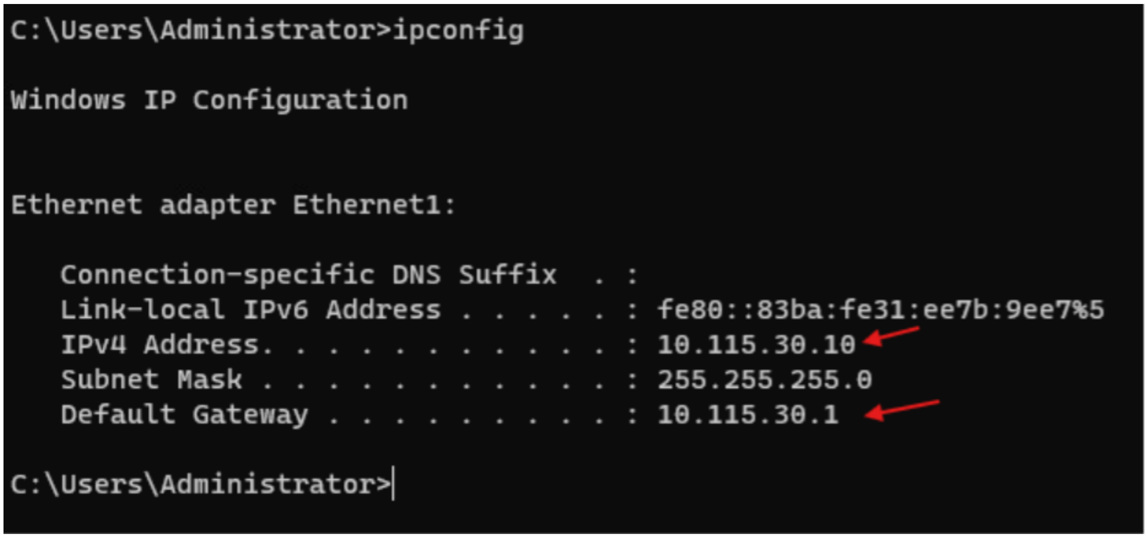


Figure 5 – Static IP configuration on the Server

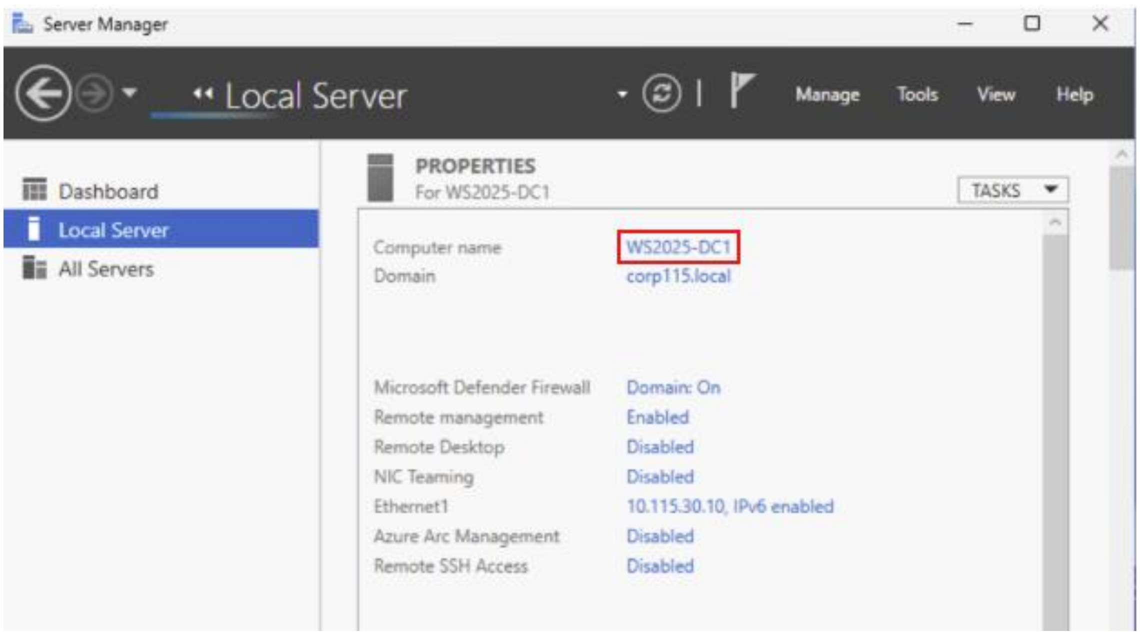


Figure 6 - Windows Server system information showing updated hostname

Part 2 – VLAN Segmentation and Network Configuration

2.1 Overview

The objective of part 2 was to design and implement segmented network architecture using VLANs. VLAN segmentation was utilized to logically separate different types of devices within the network, improving security, performance, and manageability. This phase included configuring VLANs, enabling Inter-VLAN routing, and expanding DHCP services.

2.2 VLAN Mapping Table

VLAN ID	Name	Description	Subnet
10	Classroom Clients	Physical lab workstations	10.115.10.0/24
20	Virtual Clients	VM acting as end-user **(Windows 10)**	10.115.20.0/24
30	Virtual Servers	Infrastructure servers **(Windows server 2025)**	10.115.30.0/24
40	Physical devices	Switch, Router, and ESXi Management	10.115.40.0/24
50	Parking lot	Disabled/Unused ports for security	10.115.50.0/24
99	DMZ	Isolated external-facing zone	10.115.99.0/24

2.3 Physical Switch Configuration

To implement segmentation, Six VLANs were created and named aligned with projects documentation on the switch. This prevents VLAN mismatch errors and ensures consistent traffic tagging. Access ports were configured for devices that belong to a single VLAN, while trunk ports were configured to carry traffic for multiple VLANs between network devices.

Switch Interface	**Connected Device**	**Port Mode**	**VLAN**
G5/0/1	Classroom PC **(NUC1-WS)**	Access	VLAN 10
G5/0/3	NUC2-ESXi	Trunk	VLAN 10, 20, 30, 40, 50, 99

G5/0/2	Cisco Router	Trunk	VLAN 10, 20, 30, 40, 50, 99
G5/0/4 - 52	Unused Ports	Access	VLAN 50

Technical Note: By default, the ESXi Management Network is set to **None (0),** meaning it sends untagged frames. If the switch port expects tagged traffic for VLAN 40, management connectivity would be lost. So, instead of risking a management network to restart the NUC, which could cause a disconnect or host isolation, I reconfigured the physical switch port's **Native VLAN** to **40**. While this restores connectivity, but in real network environment, implementing Virtual Switch tagging directly on the ESXi host align with standard data center practices.

```
SHAMIN-S1#show vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
10   Classroom_Client                 active    Gi5/0/1
20   Virtual_Client                   active
30   Virtual_server                   active
40   Physical_Devices                 active
50   Parking_Lot                      active    Gi5/0/4, Gi5/0/5, Gi5/0/6
                                                Gi5/0/7, Gi5/0/8, Gi5/0/9
                                                Gi5/0/10, Gi5/0/11, Gi5/0/12
                                                Gi5/0/13, Gi5/0/14, Gi5/0/15
                                                Gi5/0/16, Gi5/0/17, Gi5/0/18
                                                Gi5/0/19, Gi5/0/20, Gi5/0/21
                                                Gi5/0/22, Gi5/0/23, Gi5/0/24
                                                Gi5/0/25, Gi5/0/26, Gi5/0/27
                                                Gi5/0/28, Gi5/0/29, Gi5/0/30
                                                Gi5/0/31, Gi5/0/32, Gi5/0/33
                                                Gi5/0/34, Gi5/0/35, Gi5/0/36
                                                Gi5/0/37, Gi5/0/38, Gi5/0/39
                                                Gi5/0/40, Gi5/0/41, Gi5/0/42
                                                Gi5/0/43, Gi5/0/44, Gi5/0/45
                                                Gi5/0/46, Gi5/0/47, Gi5/0/48

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
                                                Gi5/0/49, Gi5/0/50, Gi5/0/51
                                                Gi5/0/52
99   DMZ                              active
1002 fddi-default                     act/unsup
```

Figure 7 – Displaying all the VLANs on Switch

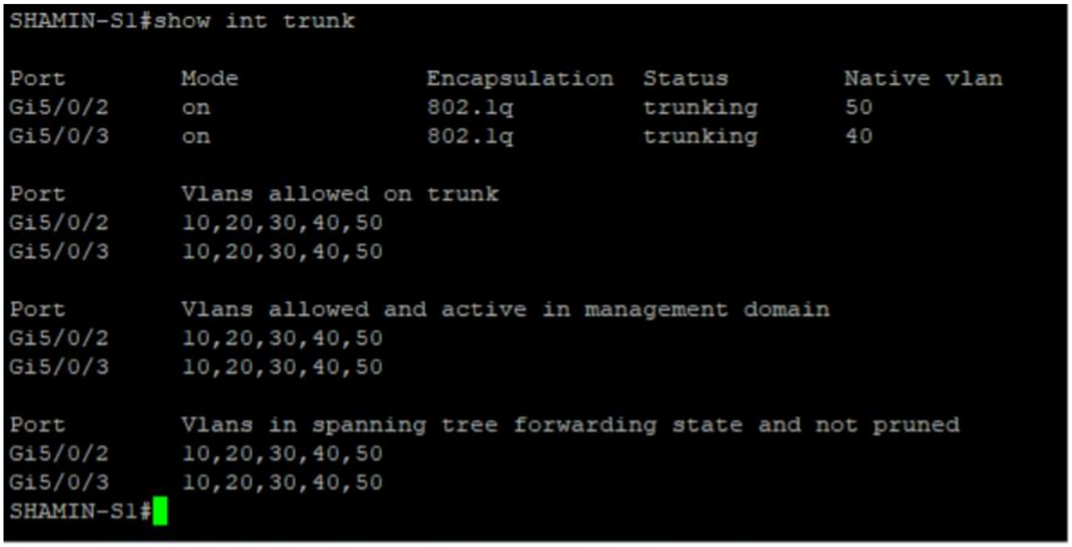


Figure 8 - Confirming Trunk configuration on Switch

2.4 ESXi Virtual Switch and VLAN Configuration

Within the VMware ESXi environment, a virtual switch **(vSwitch)** was configured to support VLAN segmentation for virtual machines. Multiple port groups were created on vSwitch, and each associated with a specific VLAN ID.

Each virtual machine was connected to the appropriate port group based on its role in the network. This ensures that virtual machines are also logically separated in the same way as physical devices.

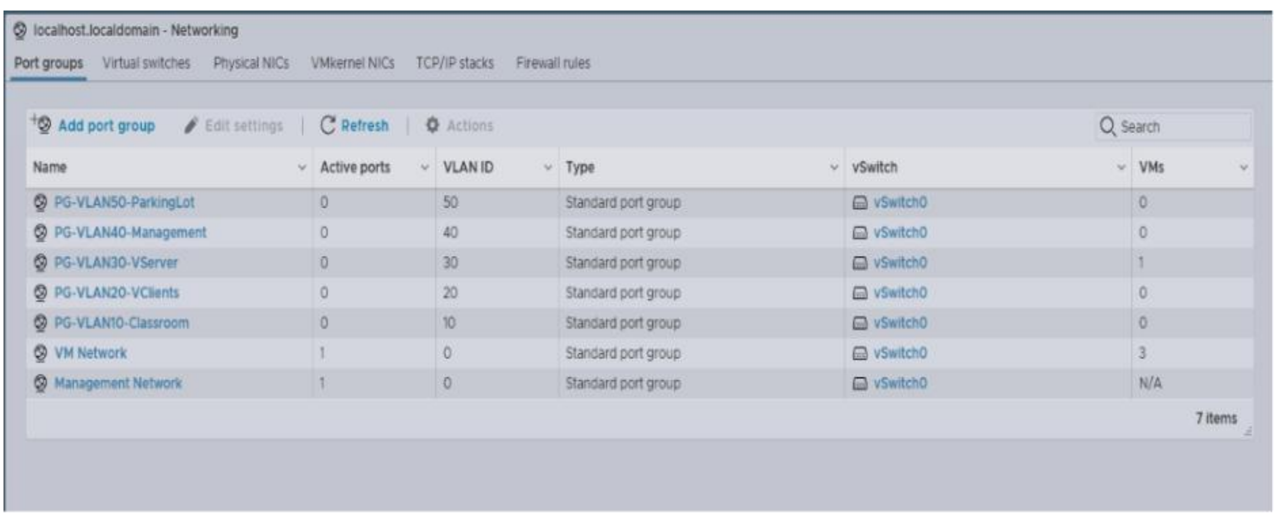


Figure 9 – ESXi vSwitch port groups showing assigned VLAN IDs

2.5 Inter-VLAN Routing Configuration

To enable communication between six isolated VLANs, inter-VLAN routing was implemented using a cisco router configured with **Router-on-a-Stick (ROAS).** This setup allows a single physical interface on the router to act as the default gateway for all virtual and physical devices by using logical sub-interfaces. Each sub-interfaces were assigned an IP address to act as the default gateway for its respective VLANs.

Subinterface	VLAN ID	IP Address	Purpose
G0/0.10	10	10.115.10.1	Classroom Clients
G0/0.20	20	10.115.20.1	Virtual Clients
G0/0.30	30	10.115.30.1	Virtual Servers
G0/0.40	40	10.115.40.1	Physical Management
G0/0.50	50	10.115.50.1	Unused Switch ports
G0/0.99	99	10.115.99.1	DMZ

```
SHAMIN-R1#show ip int br
Interface                  IP-Address      OK? Method Status                Protocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down down
GigabitEthernet0/0         unassigned      YES unset  up                    up
GigabitEthernet0/0.10      10.115.10.1     YES manual up                    up
GigabitEthernet0/0.20      10.115.20.1     YES manual up                    up
GigabitEthernet0/0.30      10.115.30.1     YES manual up                    up
GigabitEthernet0/0.40      10.115.40.1     YES manual up                    up
GigabitEthernet0/0.50      10.115.50.1     YES manual up                    up
GigabitEthernet0/0.99      unassigned      YES unset  up                    up
GigabitEthernet0/1         172.16.1.115    YES manual down                  down
Serial0/0/0                unassigned      YES unset  administratively down down
Serial0/0/1                unassigned      YES unset  administratively down down
SHAMIN-R1#
```

Figure 10 – Displaying interface G0/0 sub-interfaces on Router

2.6 External Network Connectivity (TS Network)

To provide this internal infrastructure environment with access to external resources and the management webpage at **172.16.2.100**, a dedicated connection was established to the **TS- Switch**. A separate unused router port **(G0/1)** was connected to the TS switch, and this port was configured with IP address **172.16.1.115**. A static route was added to the routing table to ensure traffic destined for the **172.16.2.0/24** network is correctly forwarded through the 172.16.1.115 gateway.

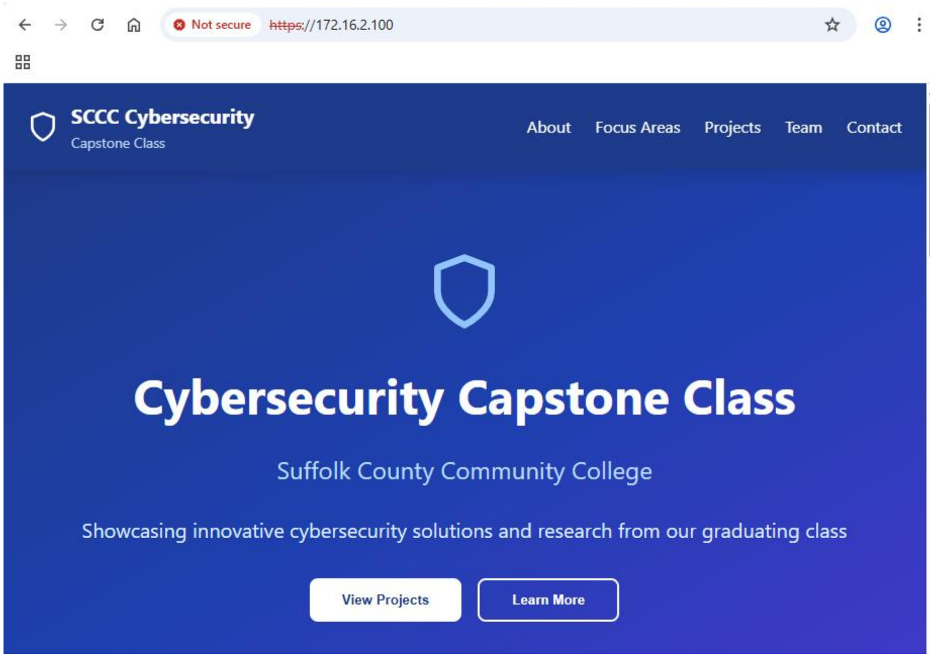


Figure 11 – Accessing the web page at 172.16.2.100

2.7 DHCP Scope Expansion

The DHCP server configured in Part 1 was expanded to support multiple VLANs by creating additional scopes for each client network.

Two DHCP scopes were created to service the client VLANs. Each scope was configured with a specific **Address Pool** to prevent IP conflicts and ensure logical organization:

- VLAN 10 (Classroom Clients): Address Pool **10.115.10.50 – 10.115.10.100**
- VLAN 20 (Virtual Clients): Address Pool **10.115.20.50 – 10.115.20.100**

This ensures that client devices automatically receive the correct network configuration based on the VLANs they are connected to.

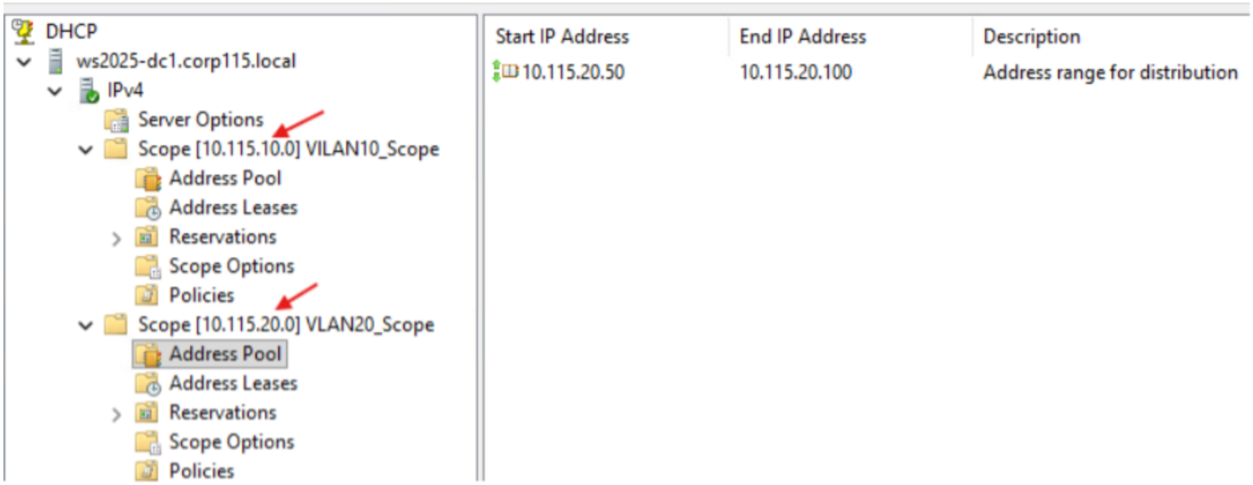


Figure 12 – DHCP console showing multiple scopes

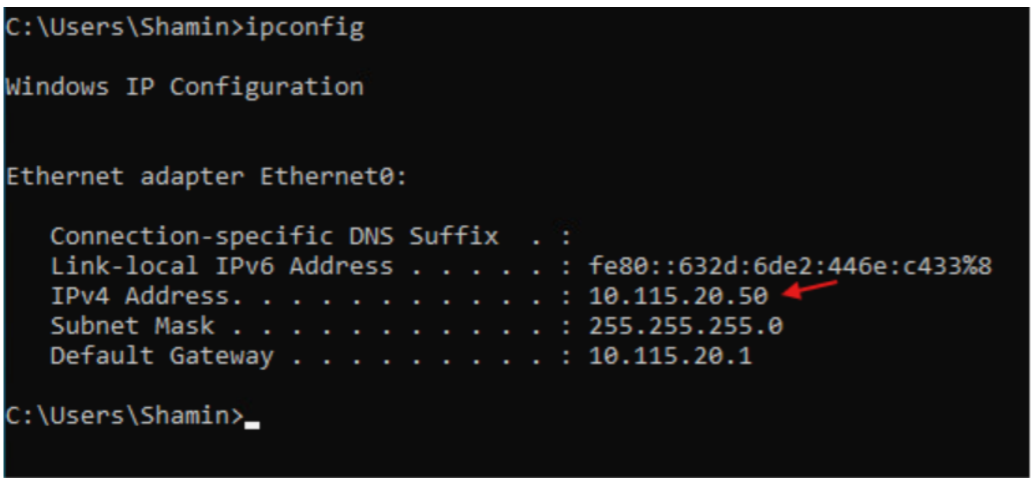


Figure 13 – Windows 10 VM receiving IP address through DHCP

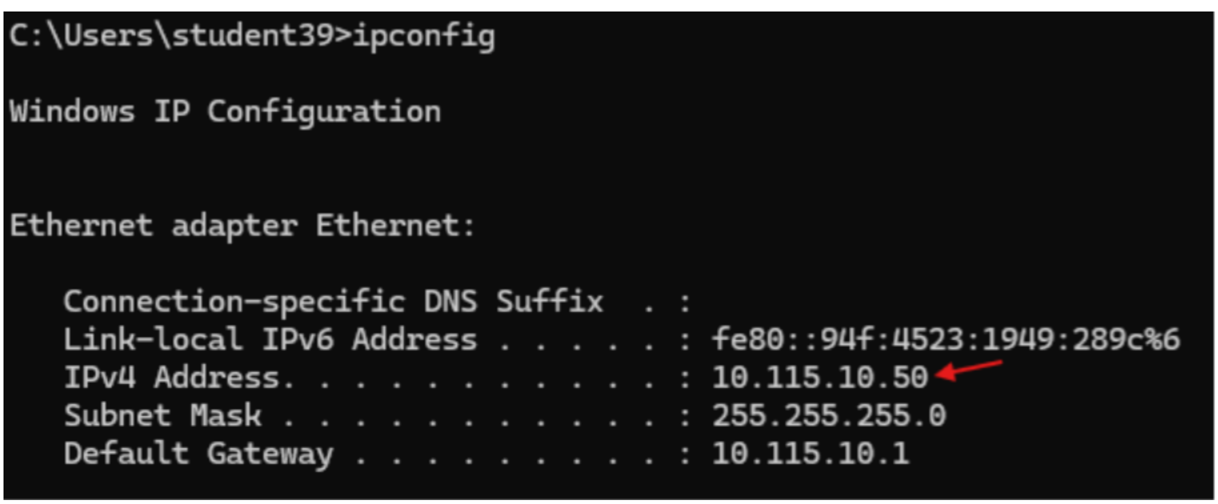
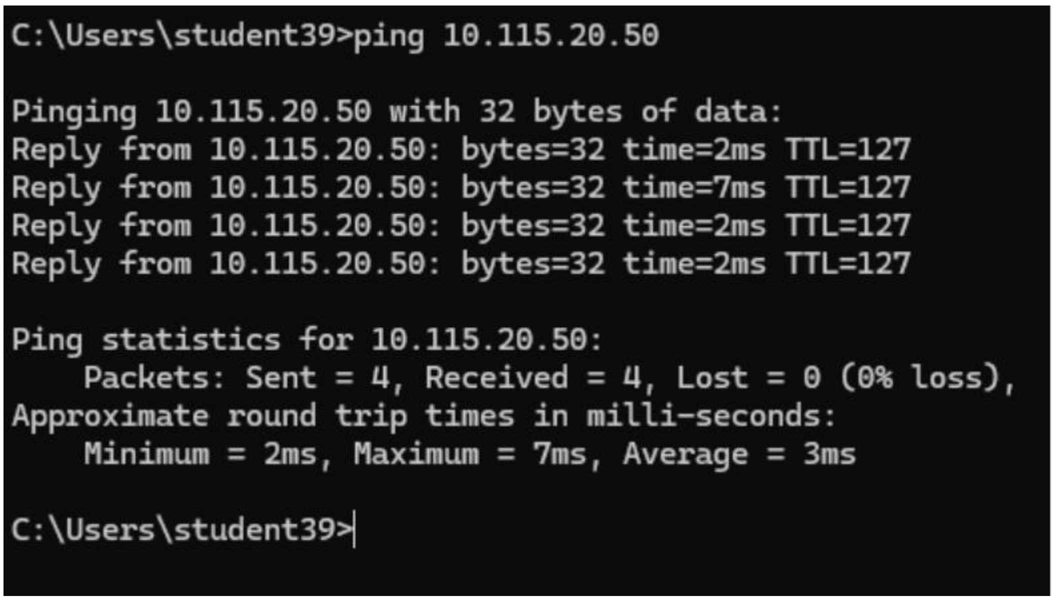


Figure 14 – Classroom computer receiving IP address from DHCP

2.8 Network Connectivity and Validation Testing

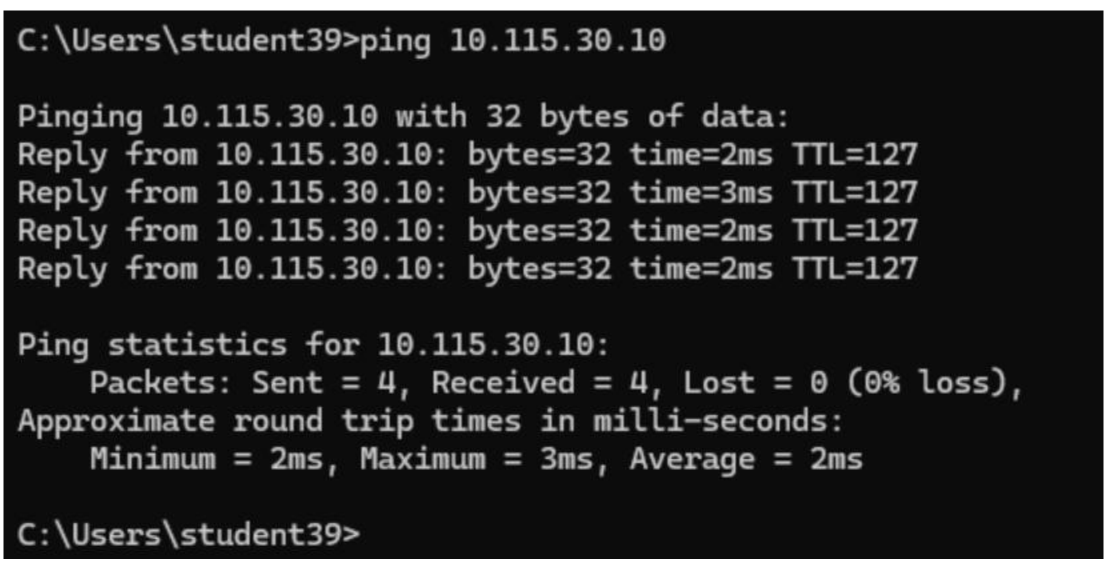
To confirm that the network is fully operational, a series of **ICMP echo requests (pings)** were performed from various endpoints:

- Perform successfully pinging between a Virtual Client **(VLAN 20)** and the Classroom PC **(VLAN 10).**
- Tested connectivity from both physical and virtual clients to Windows Server 2025 **(VLAN 30)**.



```
C:\Users\student39>ping 10.115.20.50

Pinging 10.115.20.50 with 32 bytes of data:
Reply from 10.115.20.50: bytes=32 time=2ms TTL=127
Reply from 10.115.20.50: bytes=32 time=7ms TTL=127
Reply from 10.115.20.50: bytes=32 time=2ms TTL=127
Reply from 10.115.20.50: bytes=32 time=2ms TTL=127

Ping statistics for 10.115.20.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 7ms, Average = 3ms

C:\Users\student39>
```

Figure 15 – Classroom PC to Virtual Client ping



```
C:\Users\student39>ping 10.115.30.10

Pinging 10.115.30.10 with 32 bytes of data:
Reply from 10.115.30.10: bytes=32 time=2ms TTL=127
Reply from 10.115.30.10: bytes=32 time=3ms TTL=127
Reply from 10.115.30.10: bytes=32 time=2ms TTL=127
Reply from 10.115.30.10: bytes=32 time=2ms TTL=127

Ping statistics for 10.115.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Users\student39>
```

Figure 16 – Classroom PC to Windows Server ping

2.9 Future ESXi Host VLAN Configuration Guide

I.Virtual Switch (vSwitch) Standardization:
To maintain network consistency, every new host must utilize a standard virtual switch mapped to a physical uplink that connects to the Cisco trunk fabric.

- **Navigate to Networking:** Log in to the ESXi Host Client and select **Networking** from the Navigator pane.
- **Add Standard vSwitch:** If a dedicated switch is required, select **Virtual switches** > **Add standard virtual switch**.
- **Uplink Selection:** Assign at least one physical NIC (**vmnic**) to the switch. Ensure this physical port on the NUC is cabled to a designated trunk port on the physical Cisco switch.

II. Port Group Creation & VLAN Tagging
Virtual machines do not communicate directly with the vSwitch; they connect to **Port Groups**. Each Port Group must be tagged with the correct VLAN ID to ensure the physical switch can route the traffic.

- **Add Port Group:** Under the **Port groups** tab, select **Add port group**.
- **Naming Convention:** Use the exact names established in the initial deployment such as Virtual Clients, Virtual servers.
- **VLAN ID Assignment:** Enter the specific VLAN ID such as 20 for Clients, 30 for Servers.

III. Physical Switch Trunk Verification
Before migrating production VMs to the new host, the physical switchport must be verified to ensure it is passing the required VLAN tags.

1. **Port Configuration:** Access the Cisco Switch and ensure the interface facing the new host is set to trunk mode:
    - switchport mode trunk
    - switchport trunk allowed vlan 10,20,30,40,50, 99

2. **Native VLAN Check:** If the new host's management IP is untagged, ensure the switchport trunk native VLAN command is applied to that specific interface to prevent management isolation.

Part 3 – Network Device Configuration and Secure Remote Access

3.1 Overview

The objective of Part 3 was to configure network devices, switch and router, to support secure management and proper network functionality. This phase focused on device identification, secure remote access using SSH, proper IP addressing, and securing unused ports.

3.2 Switch Configuration

The network switch was configured to support secure management and proper integration into the network infrastructure. A hostname was assigned to properly identify the device and a MOTD banner was configured to display a warning message to unauthorized user attempting to access the device.

An SVI was configured within the designated management VLAN **(VLAN 40)** to allow remote management to the switch. This interface was assigned an IP address **(10.115.40.2)** within the appropriate subnet, enabling communication with the Windows Server.

SSH access was configured to allow secure remote login to the switch. This included generating RSA keys, setting a domain name, and configuring VTY lines to accept only SSH connections.

Furthermore, all unused switch ports were administratively down to reduce potential risk and improve network security.

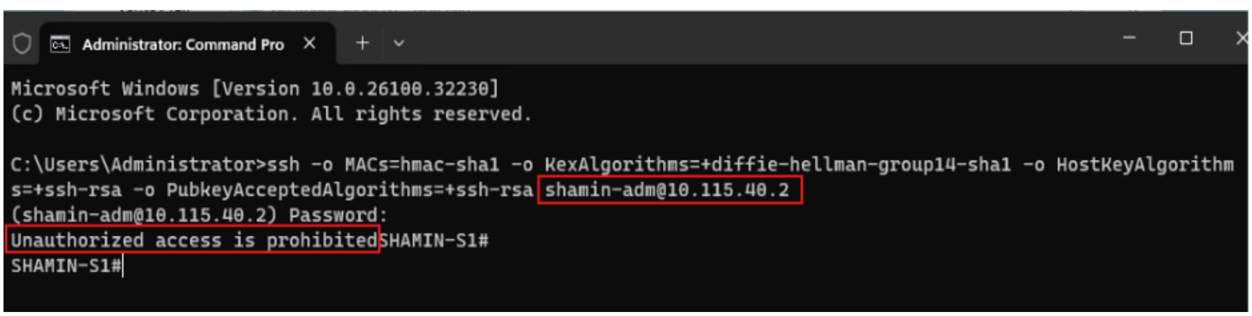


Figure 17 – SSH connection from Windows Server to Switch

3.3 Router Configuration

The cisco router was configured to support secure management and proper communication with the switch and other network devices. A hostname was assigned to identify the router, and a MOTD banner was configured to provide a security warning to unauthorized users trying to access the device.

The router interface connected to the switch was configured with an IP address within the appropriate subnet, which enables communication between the router and the rest of the network infrastructure.

SSH access was also configured on the router to allow secure remote access. Similar to switch, this included generating an RSA keys, setting a domain name, and restricting remote access to SSH only.

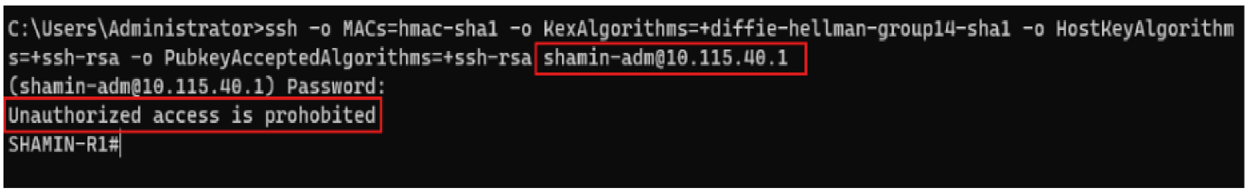


Figure 18 – SSH connection from Windows Server to Router

Part 4 – Active Directory Deployment and Domain Services

4.1 Overview

The objective of part 4 was to deploy and configure an Active Directory environment to provide centralized authentication, user management, and network resource control. This phase includes installing Active Directory Domain Services (AD DS), creating a domain and forest, deploying domain controllers, and implementing policies.

4.2 Active Directory Domain Services Installation

Active Directory Domain Services (AD DS) was installed on the primary Windows Server **(WS2025-DC1)** to establish a centralized directory service. The server was promoted to a domain controller by creating a new forest and domain. The domain name was set to **corp115.local**.

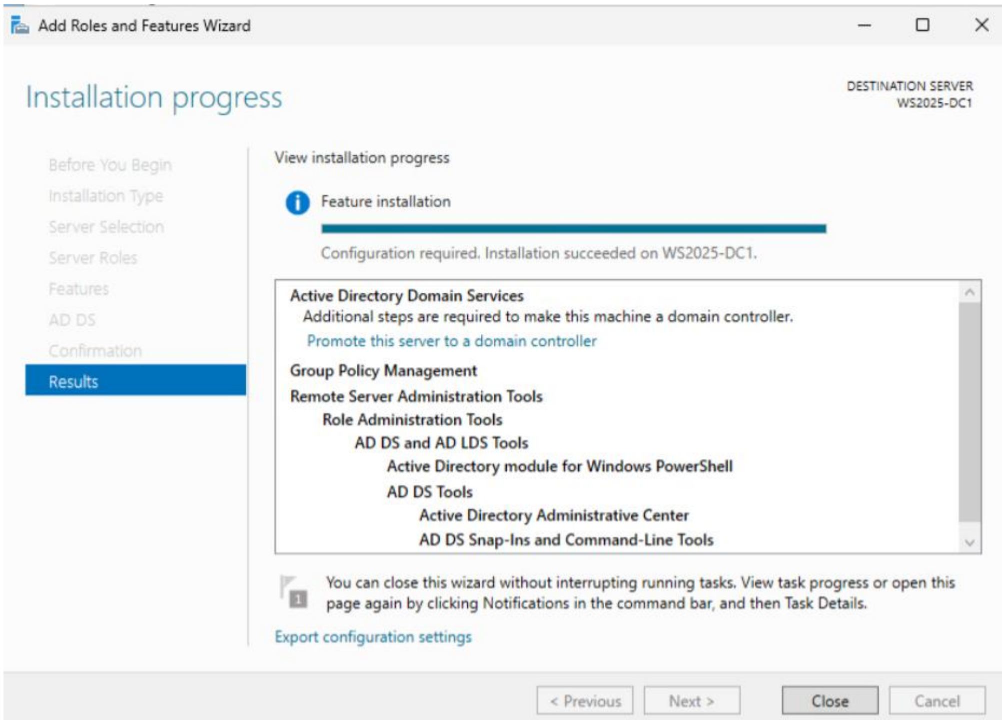


Figure 19 – AD DS role installed

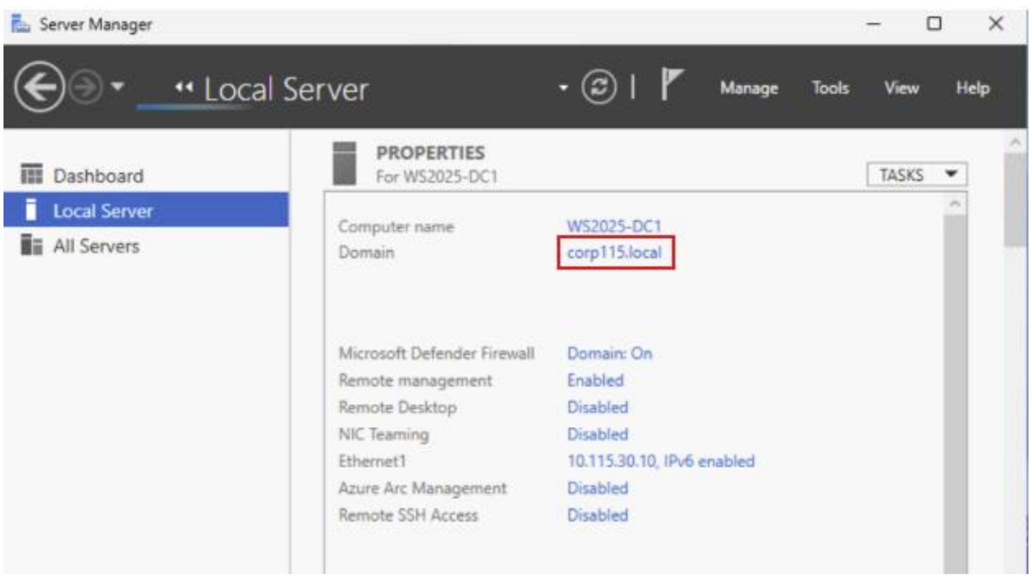


Figure 20 – Domain corp115.local

4.3 Secondary Domain Controller Deployment

To provide redundancy and fault tolerance, a second Windows Server **(WS2025-DC2)** virtual machine was deployed and joined to the domain **'corp115.local'** as another domain controller. This ensures high availability of directory services, meaning that if one domain controller fails, the directory services will remain operational.

4.4 Windows 10 Client Domain Join

A new Windows 10 **(Win10-Client)** virtual machine was created and joined to the domain **corp115.local**. This allows the client system to authenticate against Active Directory and access domain resources.

4.5 Account Configuration and Organizational Unit (OU) structure

On primary Windows server **(WS2025-DC1),** separate accounts were created for administrative and general user purposes to follow the principle of least privilege. Administrative accounts are only used for system management and user accounts are used for daily operations.

Now an organized OU was created to logically separate users and administrative accounts.

- An **IT Admin OU** was created to store administrative accounts
- An **Employee OU** was created to store general user accounts

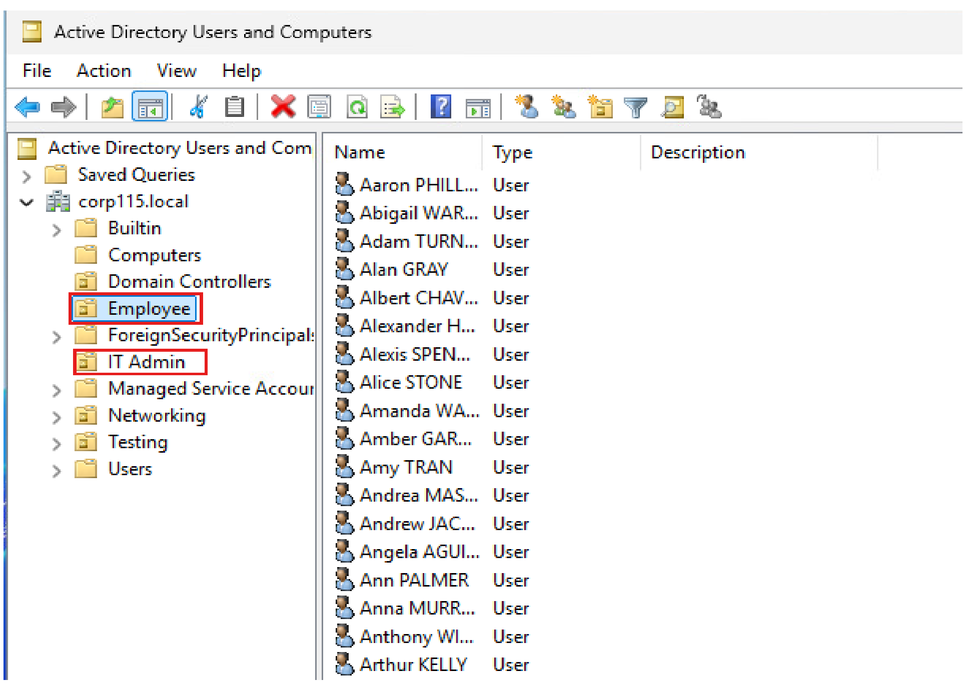


Figure 21 – ADUC showing admin and user accounts

4.6 Group Policy Configuration (Password Policy)

The domain password policy was modified to enforce stronger security requirements. Password complexity was enabled, and the minimum password length was set to 10 characters. This policy ensures that all users within the domain follow secure password practices, reducing risks of unauthorized access.

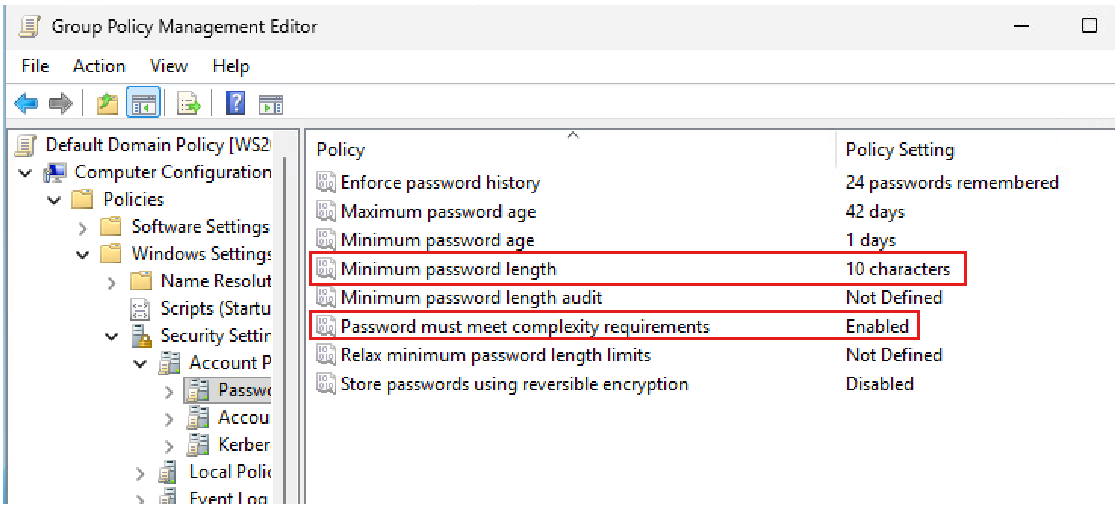


Figure 22 – Group Policy Management showing password policy setting

4.7 Bulk User Creation (Employee Accounts)

User accounts were created for employees using the provided spreadsheet. These accounts were added to the **Employee OU** to maintain organizational structure.

To create employee user accounts efficiently, the employee spreadsheet was first converted into a **CSV file**. The CSV file was required because it can import directly into PowerShell and is used to automate account creation.
Then the PowerShell script was created to read the CSV file and generate user accounts automatically in Employee OU. The script used the employee information from the CSV, such as first name, last name, and employee number, to create user objects and place them into the **Employee OU**.

```
PS C:\Users\Administrator> Import-Module ActiveDirectory

$csvPath = "C:\Users\Administrator\Documents\New Employees.csv"
$users = Import-Csv -Path $csvPath

foreach ($user in $users) {
    $first = $user.First.Trim()
    $last = $user.Last.Trim()
    $employeeID = $user.'Employee number'

    if ([string]::IsNullOrWhiteSpace($first) -or [string]::IsNullOrWhiteSpace($last)) {
        Write-Host "Skipping blank or invalid row..."
        continue
    }

    $samAccountName = "$($first.Substring(0,1))$last"
    $password = ConvertTo-SecureString "Monday123!" -AsPlainText -Force

    New-ADUser -Name "$first $last" -GivenName $first -Surname $last -SamAccountName $samAccountName

    Write-Host "Created user: $samAccountName"
}

Created user: NGONZALES
Created user: KHAWKINS
Created user: CDUNN
Created user: JJIMENEZ
Created user: PFOSTER
Created user: BROSS
Created user: LLONG
Created user: EMYERS
Created user: RPATEL
Created user: VSANDERS
Created user: ECASTILLO
Created user: CHERNANDEZ
Created user: RALVAREZ
Created user: RPRICE
Created user: RHUGHES
Created user: WRUIZ
Created user: ASPENCER
Created user: RKELLEY
Created user: JMENDOZA
Created user: AGRAY
```

Figure 23 – Execute PowerShell script to create User accounts on Employee OU

4.8 PowerShell Export of Employee OU Accounts

After employee accounts were created in Active Directory, a separate PowerShell command was used to export a list of those accounts from **Employee OU** into a text document. This export provides a simple way to document current user accounts and maintain administrative records.

```
PS C:\Users\Administrator> Get-ADUser -Filter * -SearchBase "OU=Employee,DC=corp115,DC=local" |
Select-Object Name, SamAccountName | Out-File -FilePath "C:\Users\Administrator\Documents\EmployeeList.txt" -Encoding utf8

Write-Host "Export completed successfully to C:\Users\Administrator\Documents\EmployeeList.txt"
Export completed successfully to C:\Users\Administrator\Documents\EmployeeList.txt

PS C:\Users\Administrator>
```

Figure 24 – PowerShell export Script execution

4.9 Scheduled Task Configuration

On task scheduler, a scheduled task was created to run the previously created PowerShell export command on a weekly basis. This ensures that employee account list is updated automatically without needing manual executing each time.

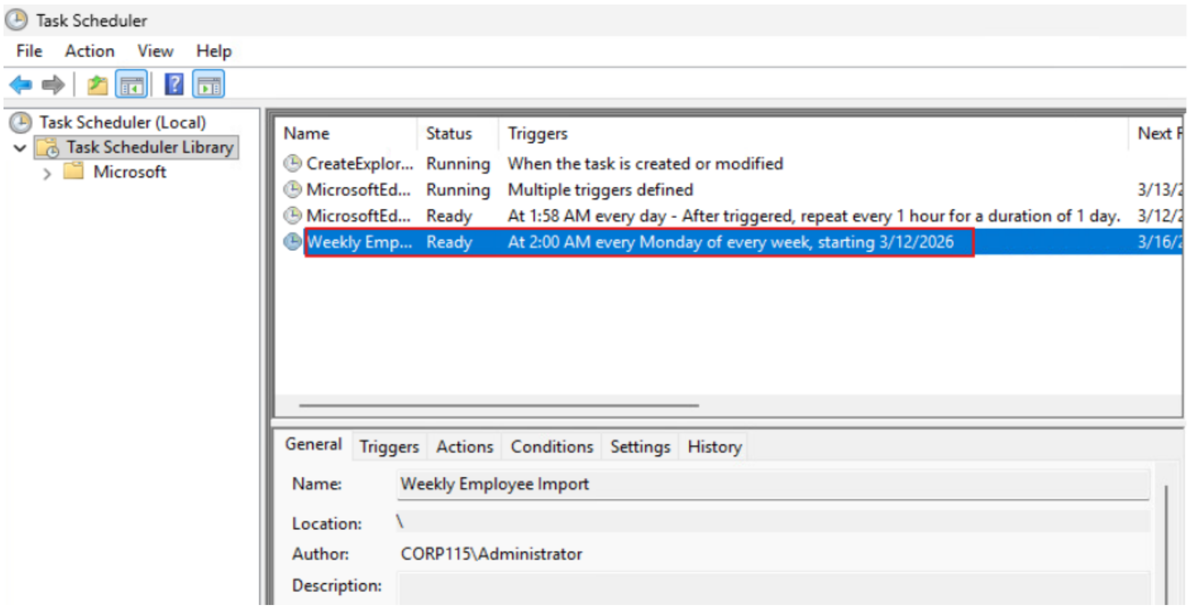


Figure 25 – Task Scheduler showing weekly task

4.10 Remote Desktop Connectivity Validation

Remote Desktop connection was tested from both client systems **(classroom PC and Windows 10 VM)** to both domain controllers. Just needed to make sure that the Remote Desktop connection is turned ON for each device.

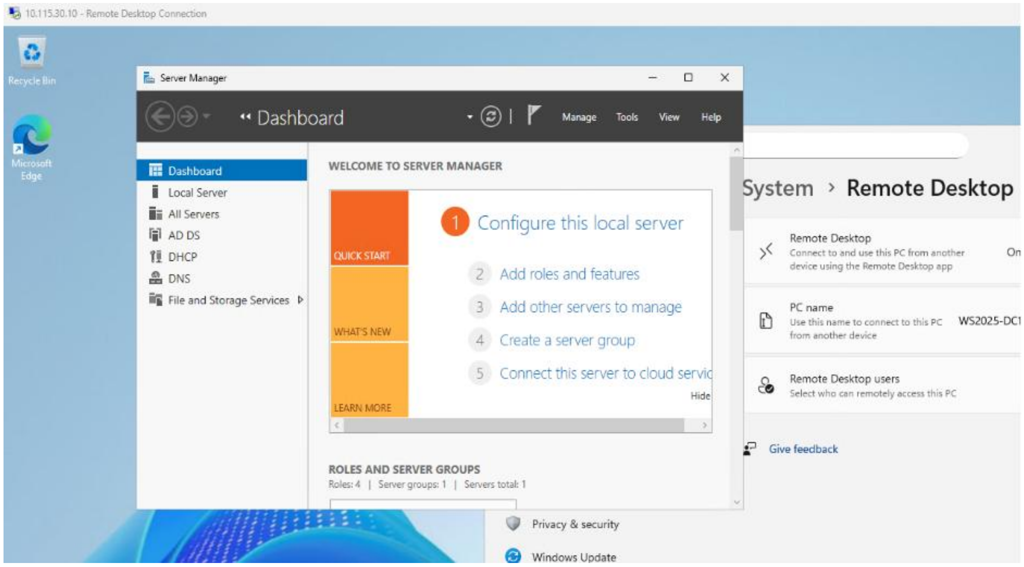


Figure 26 – RDP from classroom PC to WS2025-DC1

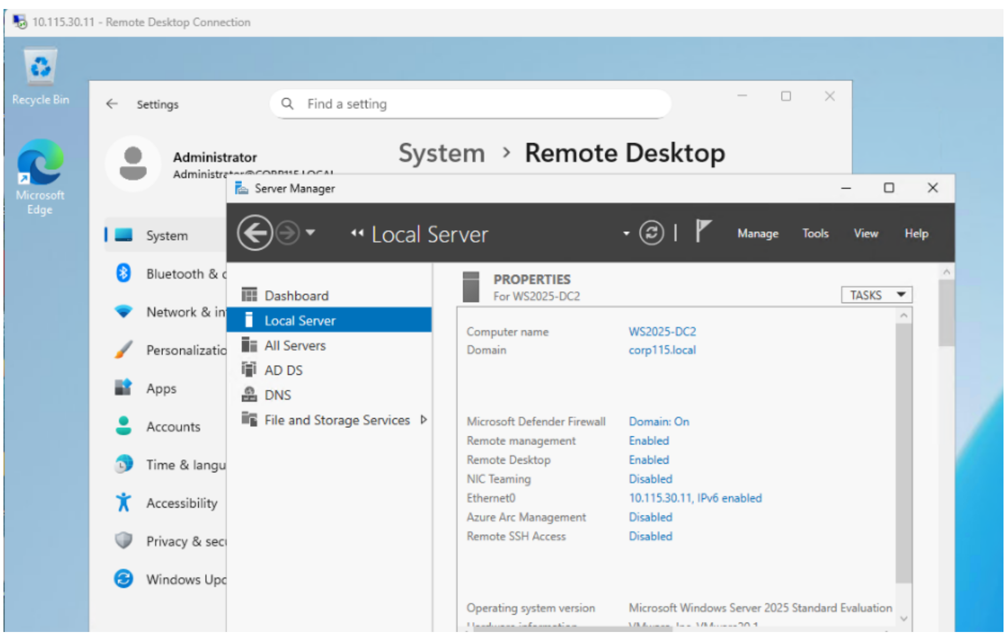


Figure 27 – RDP from Windows 10 VM to WS2025-DC2

4.11 DNS Configuration

Static DNS entries were created for all non-windows devices within the network. This ensures that network devices such as routers and switches can be resolved by hostname instead of IP address.

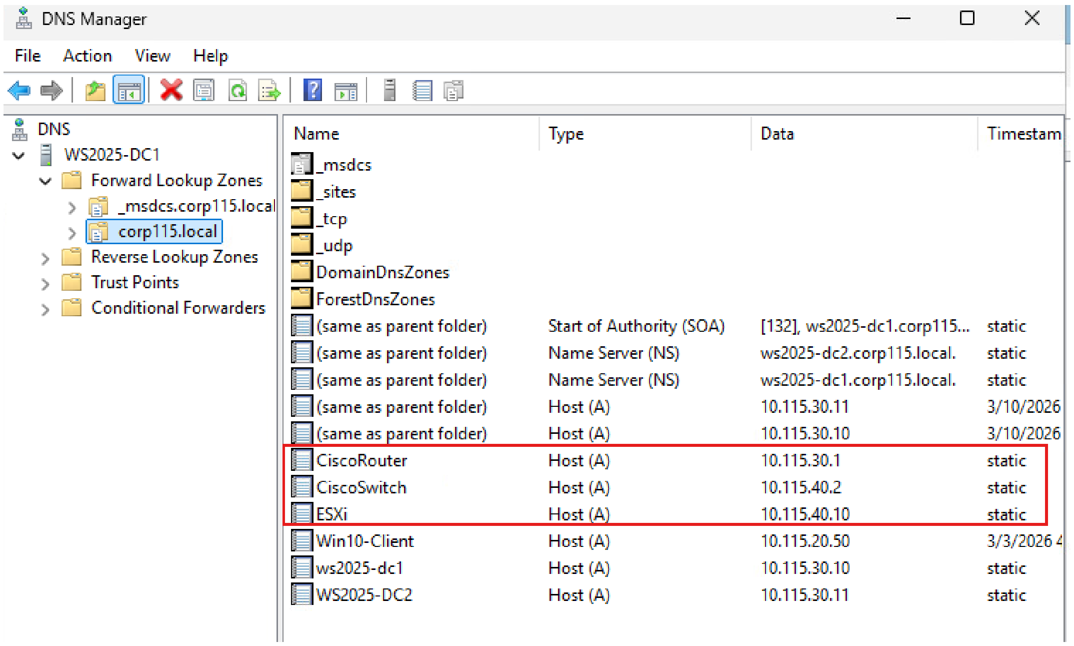


Figure 28 – DNS Manager showing static A records

4.12 Domain Controller Deployment Checklist

A standard checklist was created to guide future deployment of domain controllers. This includes:

Pre-Installation Requirements:

- Install Windows Server on the new VM.
- Assign the static IP address within the server VLAN
- Set the primary DNS to the IP of the existing Domain Controller.
- Rename the Server according to naming conventions
- Verified the network connectivity to the current domain controller

Domain Controller Installation Checklist

1. Install **Windows Server 2025** on the new machine.
2. Assign a **static IP address** to the server.
3. Configure the **Preferred DNS server** to point to the existing Domain Controller (**WS2025-DC1**).
4. Rename the server according to the organization's naming standard (for example: **WS2025-DC2**).
5. Restart the server after renaming.
6. Join the server to the existing domain **(corp115.local)**

7. Log in using a **domain administrator account**.
8. Open **Server Manager**.
9. Select **Add Roles and Features**.
10. Install the **Active Directory Domain Services (AD DS)** role.
11. After installation, click **Promote this server to a domain controller**.
12. Select **Add a domain controller to an existing domain**.
13. Enter the domain administrator credentials.
14. Select the appropriate domain **(corp115.local)**
15. Enable the following options:
    Domain Name System (DNS) Server
    Global Catalog
16. Set the **Directory Services Restore Mode (DSRM) password**.
17. Continue through the configuration wizard and confirm the installation.
18. Allow the server to restart automatically after promotion

Post Installation Verification

- The server appears in **Active Directory Users and Computers** under Domain controllers.
- DNS records are created correctly.
- Run **repadmin /showrepl** in PowerShell to confirm successful replication between domain controllers.
- Confirm that the **SYSVOL** and **Netlogon** shares are present
- Make sure the new domain controller can authenticate domain users.

4.13 Procedure for Bulk User Creation

A procedure document was created to allow lower-level administrators to create multiple employee user accounts using a CSV file and PowerShell script. This process allows administrators to quickly add large numbers of users without manually creating each account.

- **Preparing the CSV File:** First, the employee spreadsheet containing user information was converted into CSV file format. The spreadsheet included the required user information such as first name, last name, and employee number. The file was saved as CSV file so that it could be imported directly into PowerShell.

- **Importing the file using PowerShell:** Next, the CSV file was placed on the domain controller in the location referenced by the PowerShell import script. The script was designed to read each row of the CSV file and create a corresponding user account in Active Directory.

```
Import-Module ActiveDirectory

$csvPath = "C:\Users\Administrator\Documents\New
Employees.csv"
$users = Import-Csv -Path $csvPath

foreach ($user in $users) {
    $first = $user.First.Trim()
    $last = $user.Last.Trim()
    $employeeID = $user.'Employee number'

    if ([string]::IsNullOrWhiteSpace($first) -or
[string]::IsNullOrWhiteSpace($last)) {
        Write-Host "Skipping blank or invalid row..."
        continue
    }

    $samAccountName = "$($first.Substring(0,1))$last"
    $password = ConvertTo-SecureString "Monday123!" -
AsPlainText -Force

    New-ADUser -Name "$first $last" -GivenName $first
-Surname $last -SamAccountName $samAccountName -
EmployeeID $employeeID -Path
"OU=Employee,DC=corp115,DC=local" -AccountPassword
$password -ChangePasswordAtLogon $true -Enabled $true

    Write-Host "Created user: $samAccountName"
}
```

- **Creating User accounts in the correct OU**: PowerShell was then opened with administrative privileges on the domain controller. The previously created script was executed, which imported the CSV file and automatically generated user accounts based on the data in the file. During execution, the script assigned attributes such as the user's name, username, and employee ID, and placed each account in the **Employee OU.**

- **Verify account creating:** After executing the script, the results were verified using Active Directory Users and Computers to confirm that all employee accounts were successfully created in the correct OU.

Part 5 – RADIUS Authentication and Network Access Control

5.1 Overview

The objective of part 5 was to implement centralized authentication for network devices using Radius protocol. Radius allows network devices such as routers and switches to authenticate users against centralized server rather than using only local credentials.

In this network environment, Radius was installed and configured on the secondary domain controller **(WS2025-DC2).** Network devices were then configured to authenticate administrators using the RADIUS server. Role-based access control was also implemented to provide different privilege levels for **network engineers** and **network technicians.**

5.2 Radius Server Installation

The Network Policy Server (NPS) role, which provides Radius functionality in Windows Server, was installed on the second domain controller. After installation, the server was registered in Active Directory to allow it to authenticate domain users.

The router and switch were then added to the Radius server as network clients, allowing them to communicate with the server for authentication requests.

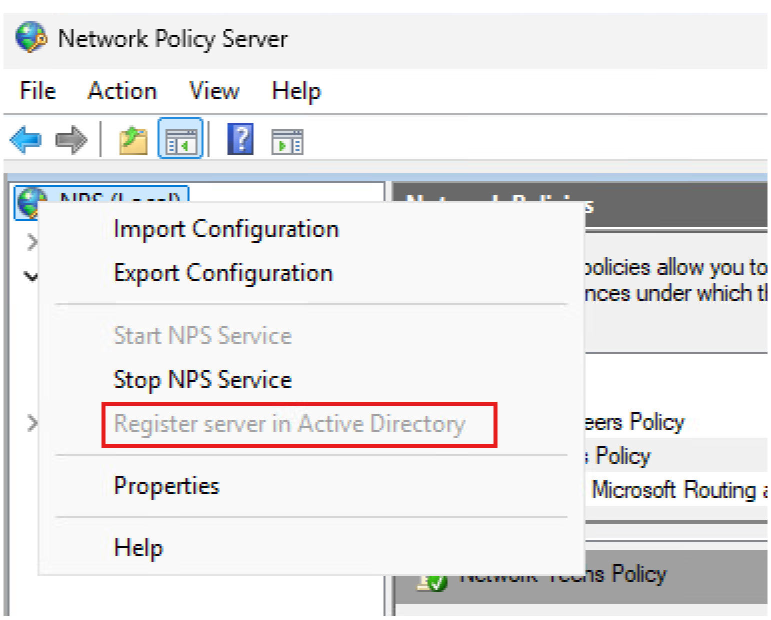


Figure 28 – NPS server installed and registered in Active Directory

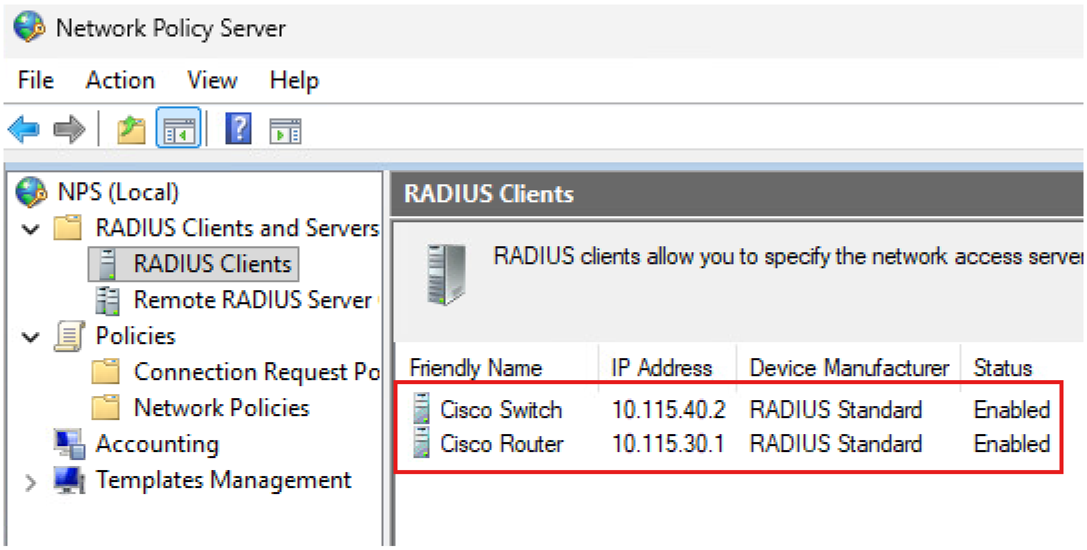


Figure 29 - RADIUS clients configured for Cisco router and switch

5.3 Active Directory Structure for Network Administration

To organize network administration roles, a **Networking organizational unit (OU)** was created within active directory. Within the Networking OU, two groups were created to represent different administrative roles:

- **Network Engineers**
- **Network Techs**

Now after creating the group, two user accounts were created to represent each administrative role. One account was assigned to the **Network Engineers** groups, and the other account was assigned to the **Network Techs** group. This structure allows administrators to manage permissions more efficiently by assigning rights to groups rather than individual users.

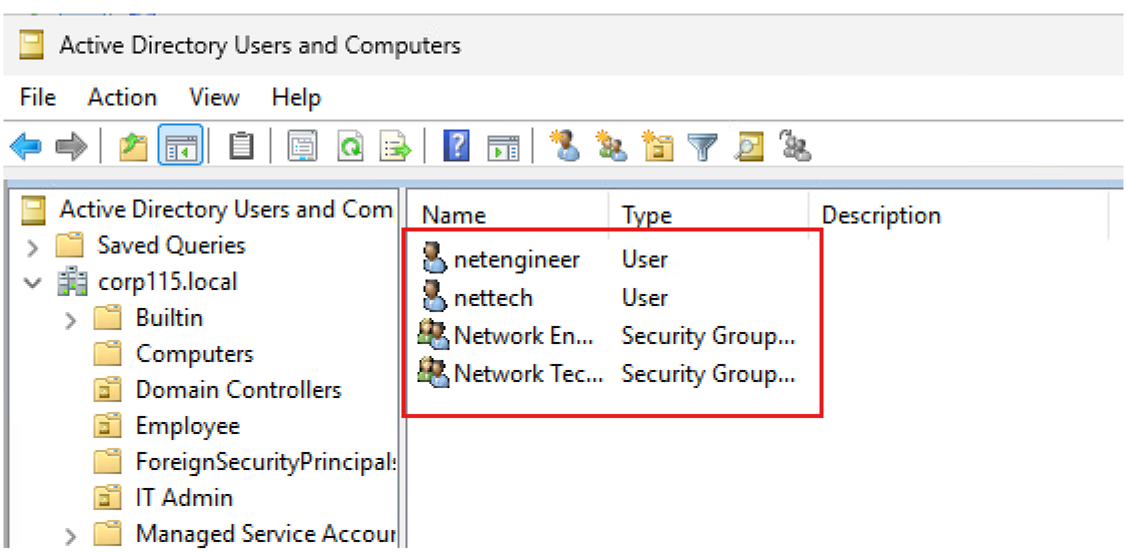


Figure 30 – Networking OU and both groups

5.4 Network Policy Server (NPS) Configuration

After installing the Network Policy Server role, the Radius service was configured to authenticate network device administrators using active directory credentials. First, the router and switch were added as Radius clients and each of these devices was configured with its management IP address and a shared secret key that matches the configuration on the network devices.

Next, **network policies** were created to control authorization for different administrative roles. Two policies were created to correspond with active directory security groups created earlier:

- **Network Engineers Policy**
- **Network Techs Policy**

Vendor-specific attributes were then applied within the policy settings to define privilege levels for Cisco devices. The Network Engineers policy was configured to grant full administrative privileges, while the Network Techs policy was configured with limited read-only access.

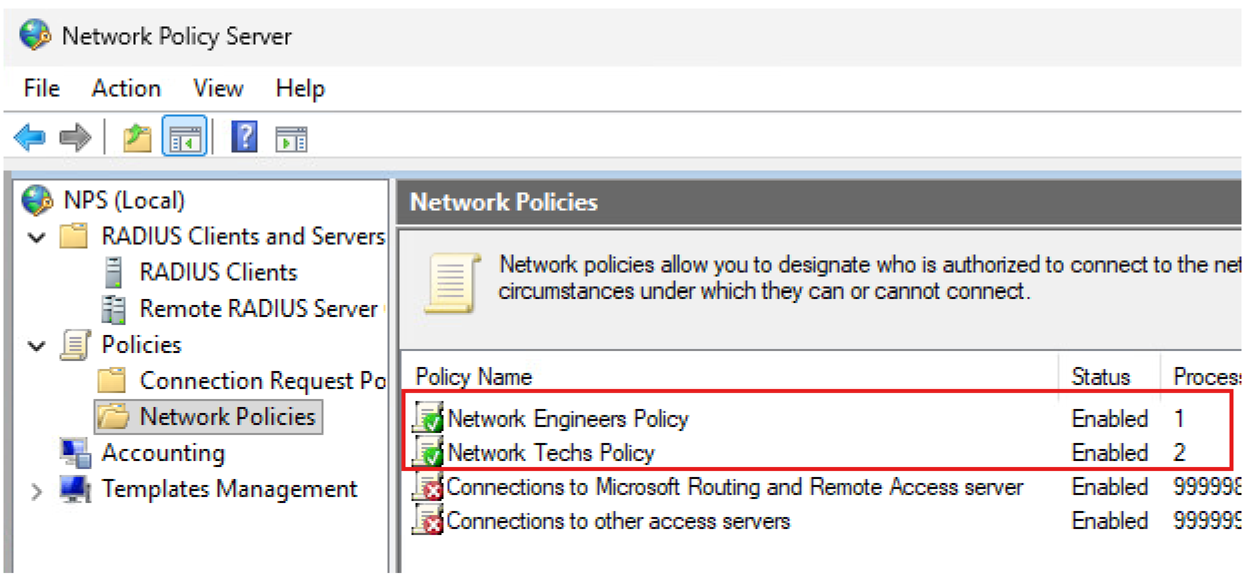


Figure 31 – NPS network policies configuration

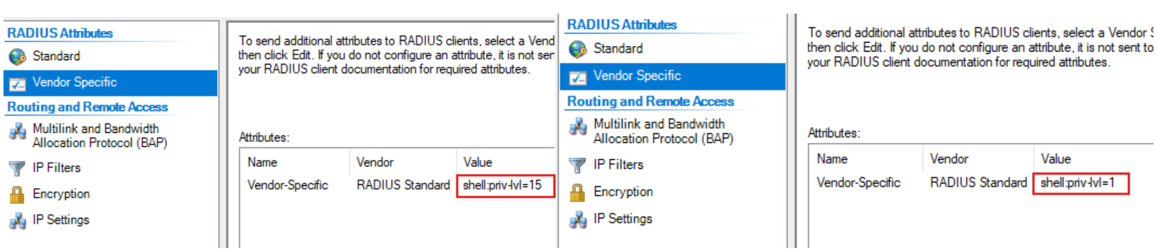


Figure 32 - Vendor-specific attribute configuration (shell:priv-lvl)

5.5 RADIUS Authentication Configuration on Network Devices

The router and switch were configured to authenticate administrative logins using the Radius server. This includes the Radius server username, shared secret key, and enabling AAA authentication on the devices.

When administrators attempt to log in through SSH, the network devices forward authentication requests to the Radius server. The server verifies the credentials against active directory and allows proper authorization for specific accounts.

```
SHAMIN-S1#show run | include aaa
aaa new-model
aaa group server radius RAD-GROUP
aaa authentication login default group RAD-GROUP local
aaa authentication login CONSOLE local
aaa authorization exec default group RAD-GROUP local
aaa session-id common
SHAMIN-S1#show run | include radius
aaa group server radius RAD-GROUP
radius server WS2025-DC2
```

```
SHAMIN-R1#show run | include aaa
aaa new-model
aaa group server radius RAD-GROUP
aaa authentication login default group RAD-GROUP local
aaa authentication login CONSOLE local
aaa authorization exec default group RAD-GROUP local
aaa session-id common
SHAMIN-R1#show run | include radius
aaa group server radius RAD-GROUP
radius server WS2025-DC2
SHAMIN-R1#
```

Figure 33 – Both Router and Switch Radius configuration

5.6 Role-Based Access Verification

Role-based access control was implemented to ensure that each administrative role receives the appropriate level of access.

- The Network Engineer group account was granted full administrative privileges on network devices
- The Network Tech group account was restricted to read-only access

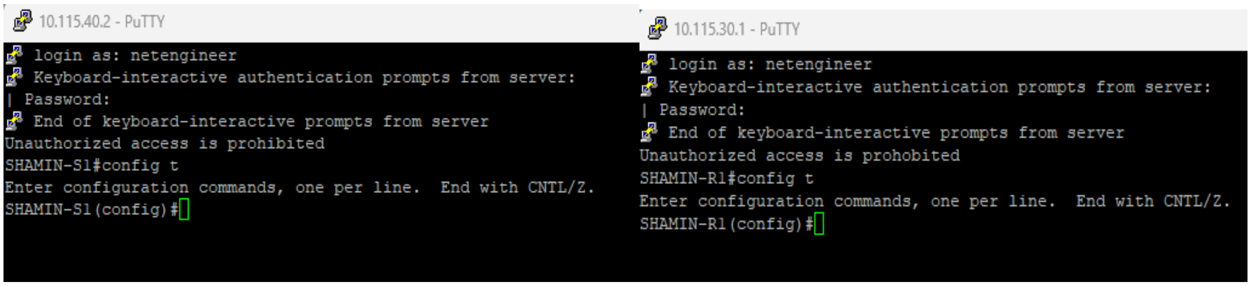


Figure 34 - SSH login as Network Engineer (full access) to both router and switch

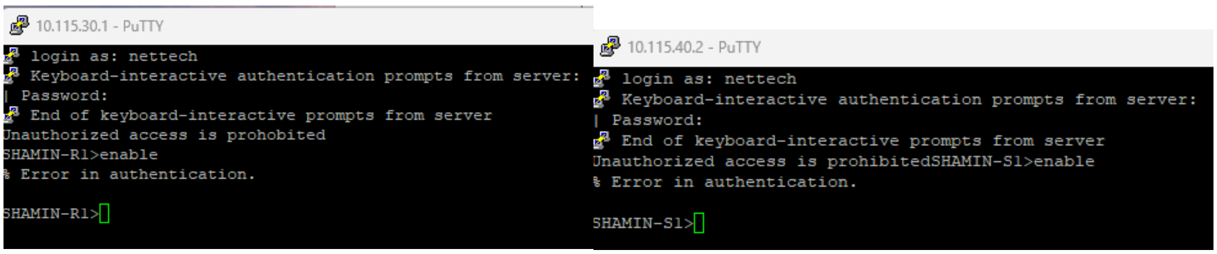


Figure 35 - SSH login as Network Tech (read-only access) to both router and switch

5.7 Verification Using PowerShell

PowerShell was used to verify the properties of the two domain user accounts created for network administration. This step confirms that the accounts exist in Active Directory and are associated with the correct groups.

.

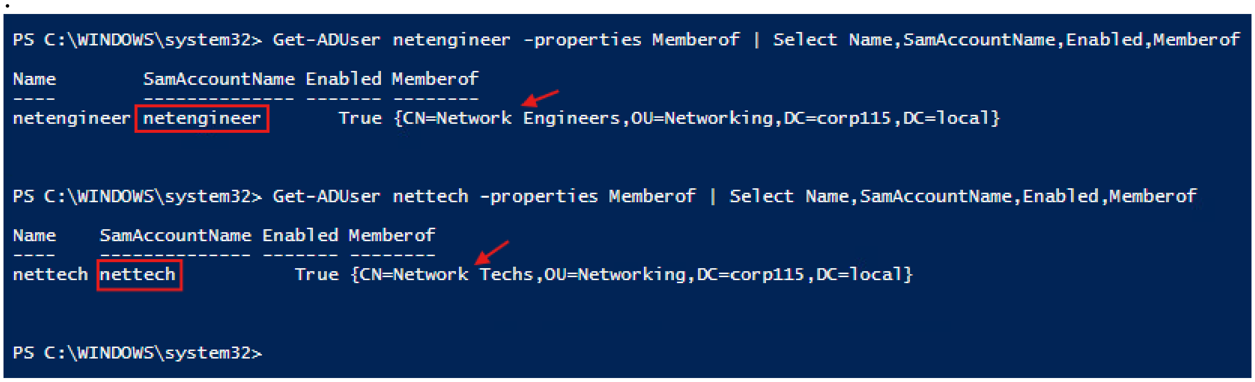


Figure 36 - PowerShell output showing properties of both user accounts

Part 6 – DMZ Implementation, Dynamic Routing, and Access Control

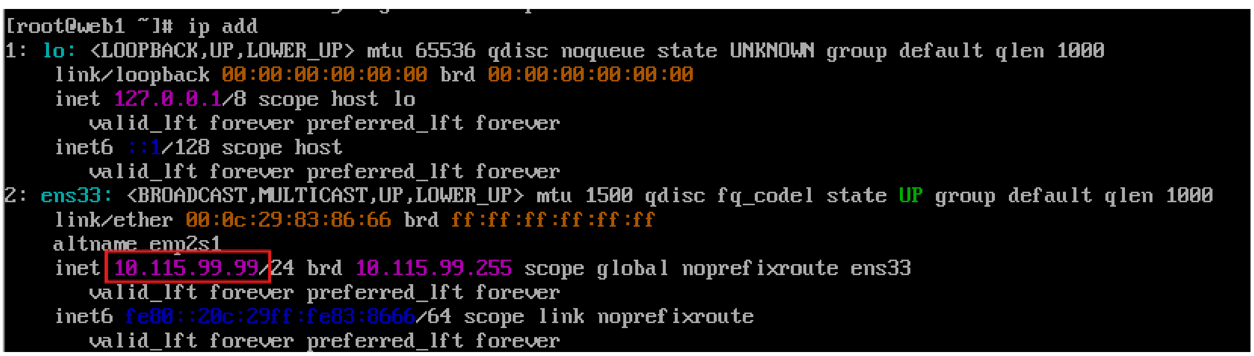
6.1 Overview

The objective of Part 6 was to implement a secure and scalable network design by introducing a DMZ (Demilitarized Zone), a CentOS based web server deployed in the DMZ VLAN, enabling dynamic routing using OSPF to advertise internal networks, and applying access control using ACLs to restrict access on source networks.

6.2 DMZ VLAN 99 and Web Server Configuration

A **DMZ VLAN (VLAN 99)** was created using the designated subnet. A provided CentOS virtual appliance was imported into the ESXi environment and configured with a static IP address in the DMZ network.

The server was used to host a custom web page that includes the administrator's name. A self-signed certificate was configured to enable **HTTPS** access to the website.



```
[root@web1 ~]# ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:83:86:66 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 10.115.99.99/24 brd 10.115.99.255 scope global noprefixroute ens33
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe83:8666/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Figure 37 – Configuring static IP on DMZ

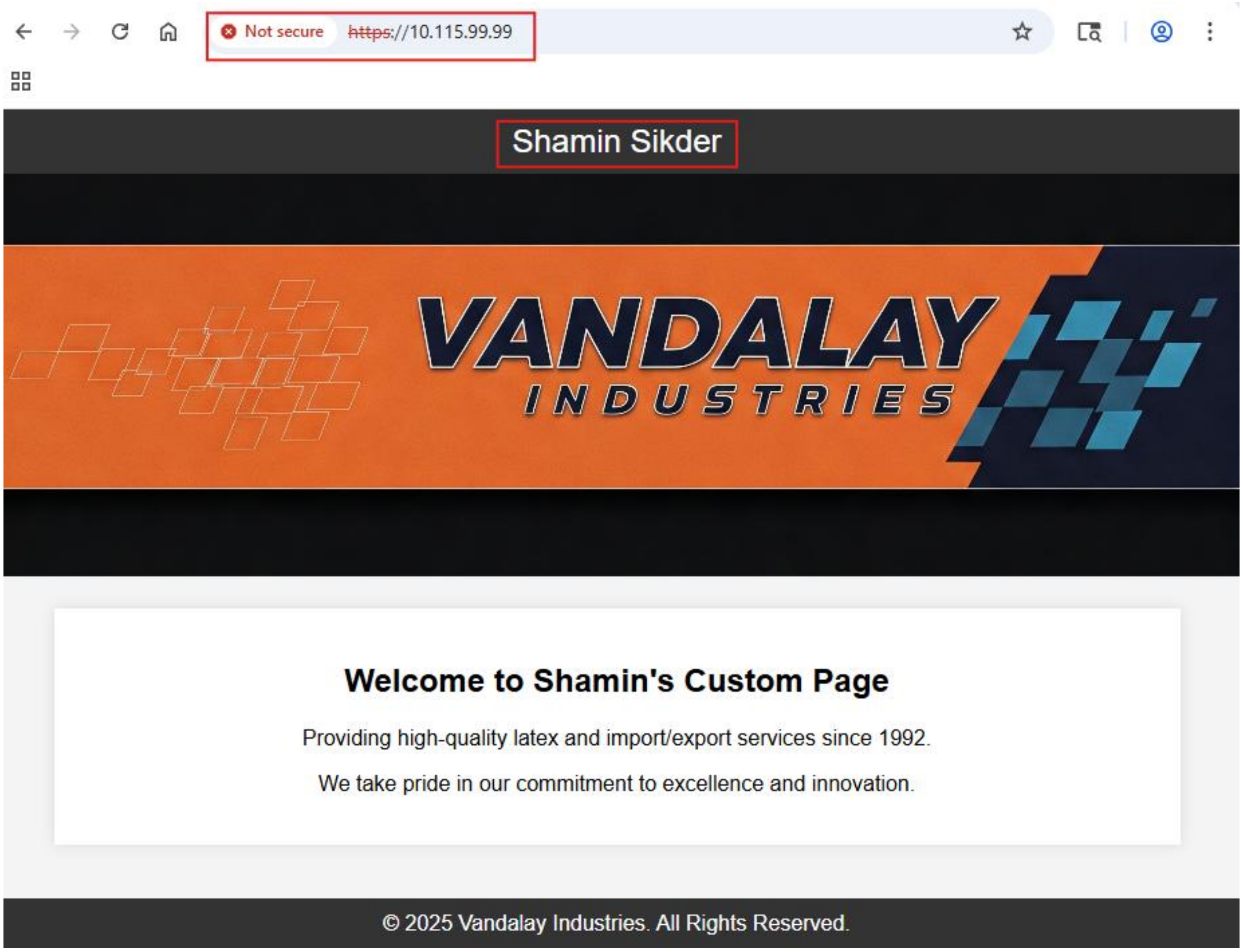


Figure 38 - Browser showing custom website with my name

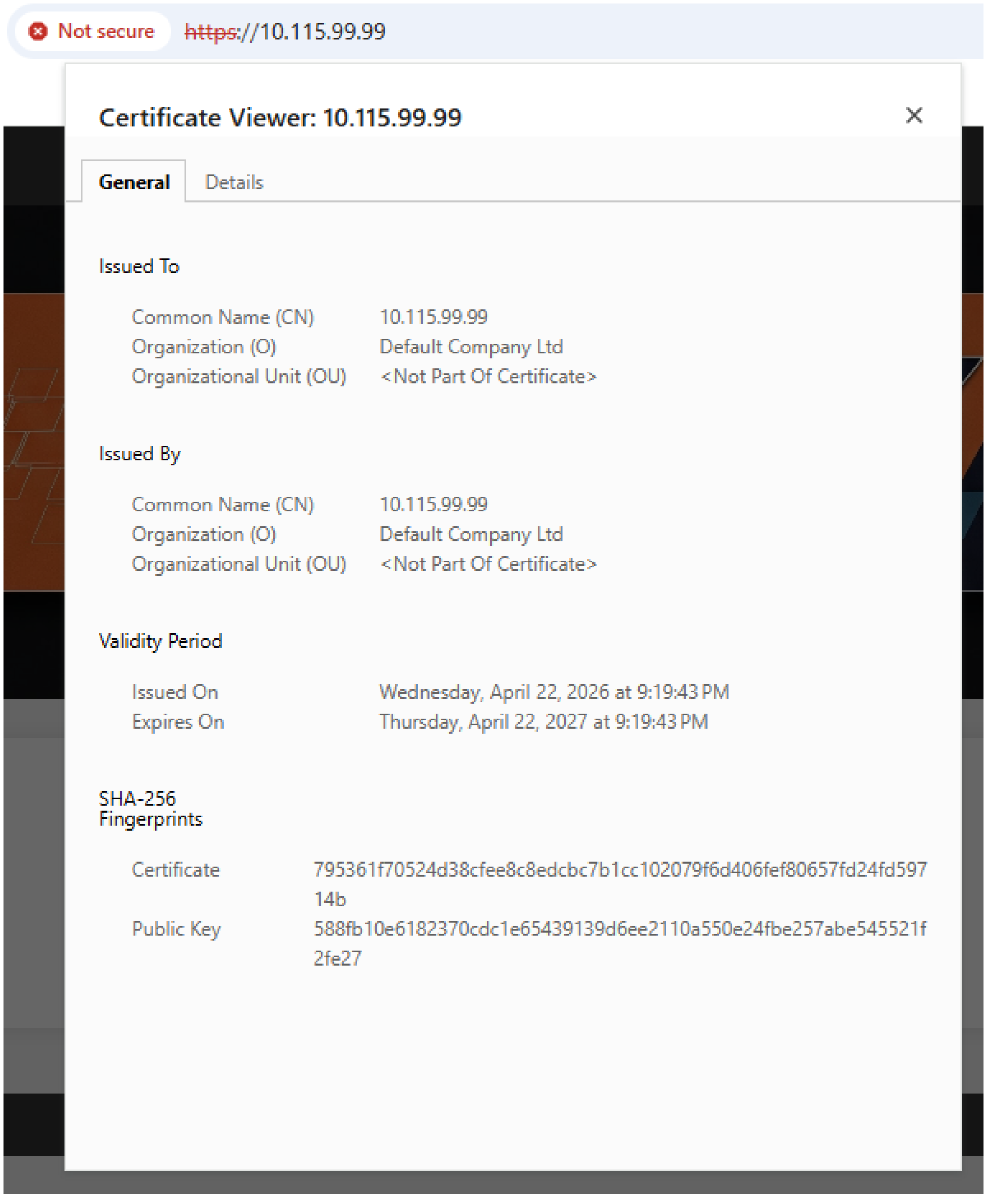
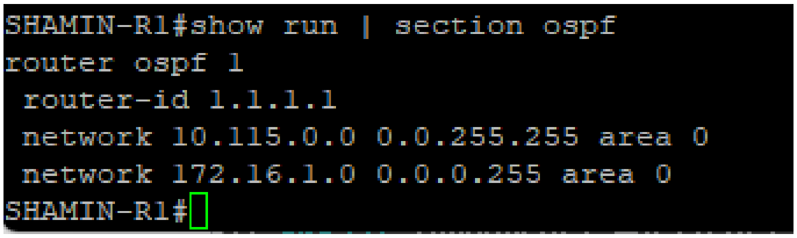


Figure 39 – Enabling HTTPs and certificate warning on the custom website

6.3 OSPF Configuration

OSPF was configured on the router in **Area 0** to dynamically advertise all internal networks. This allows routers to automatically share routing information without relying solely on static routes. The administrative distance of static routes was adjusted to be lower than OSPF routes.



```
SHAMIN-R1#show run | section ospf
router ospf 1
 router-id 1.1.1.1
 network 10.115.0.0 0.0.255.255 area 0
 network 172.16.1.0 0.0.0.255 area 0
SHAMIN-R1#
```

Figure 40 – Router configuration showing OSPF setup

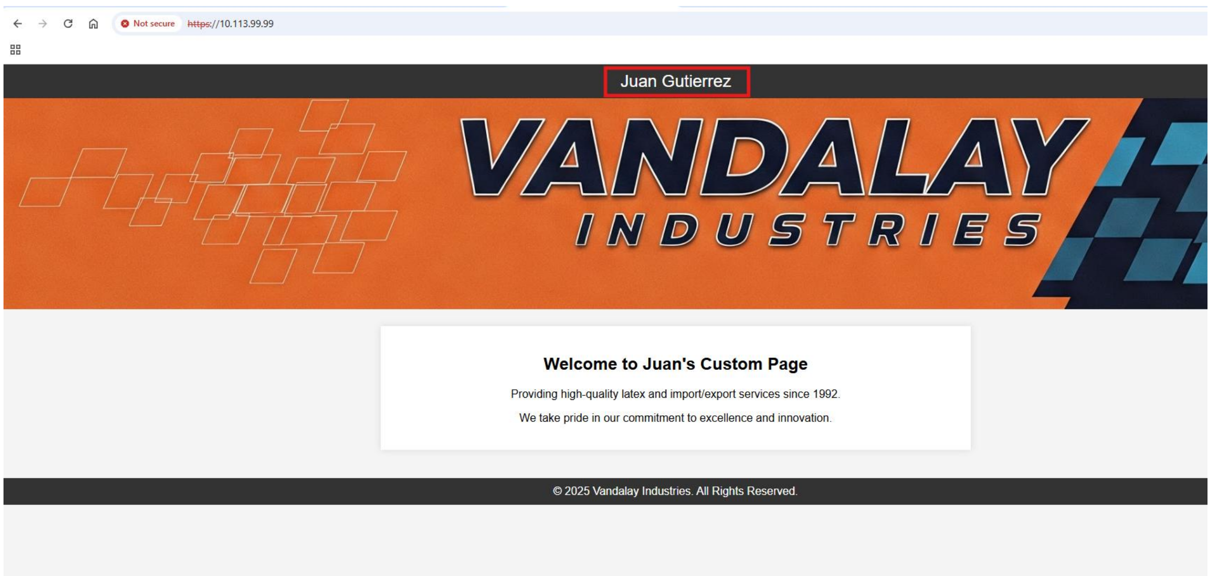


Figure 41 – Accessing other students custom websites through OSPF

6.4 Access Control List (ACL) Configuration

Access Control Lists (ACLs) were configured to control traffic between internal and external networks.

- The **TS network (172.16.2.0)** was allowed full access to the entire environment
- Other student networks were restricted to accessing only the **DMZ web server**

```
Extended IP access list DMZ-INBOUND
    10 permit ip 172.16.2.0 0.0.0.255 any (9784 matches)
    20 permit tcp any host 10.115.99.99 eq www (371 matches)
    25 permit tcp any host 10.115.70.10 eq www (105 matches)
    30 permit tcp any host 10.115.99.99 eq 443 (180 matches)
    35 permit tcp any host 10.115.70.10 eq 443
    40 deny ip any 10.115.0.0 0.0.255.255
    50 permit ip any any (1851358 matches)
SHAMIN-R1#
```

Figure 42 - ACL configuration on Router

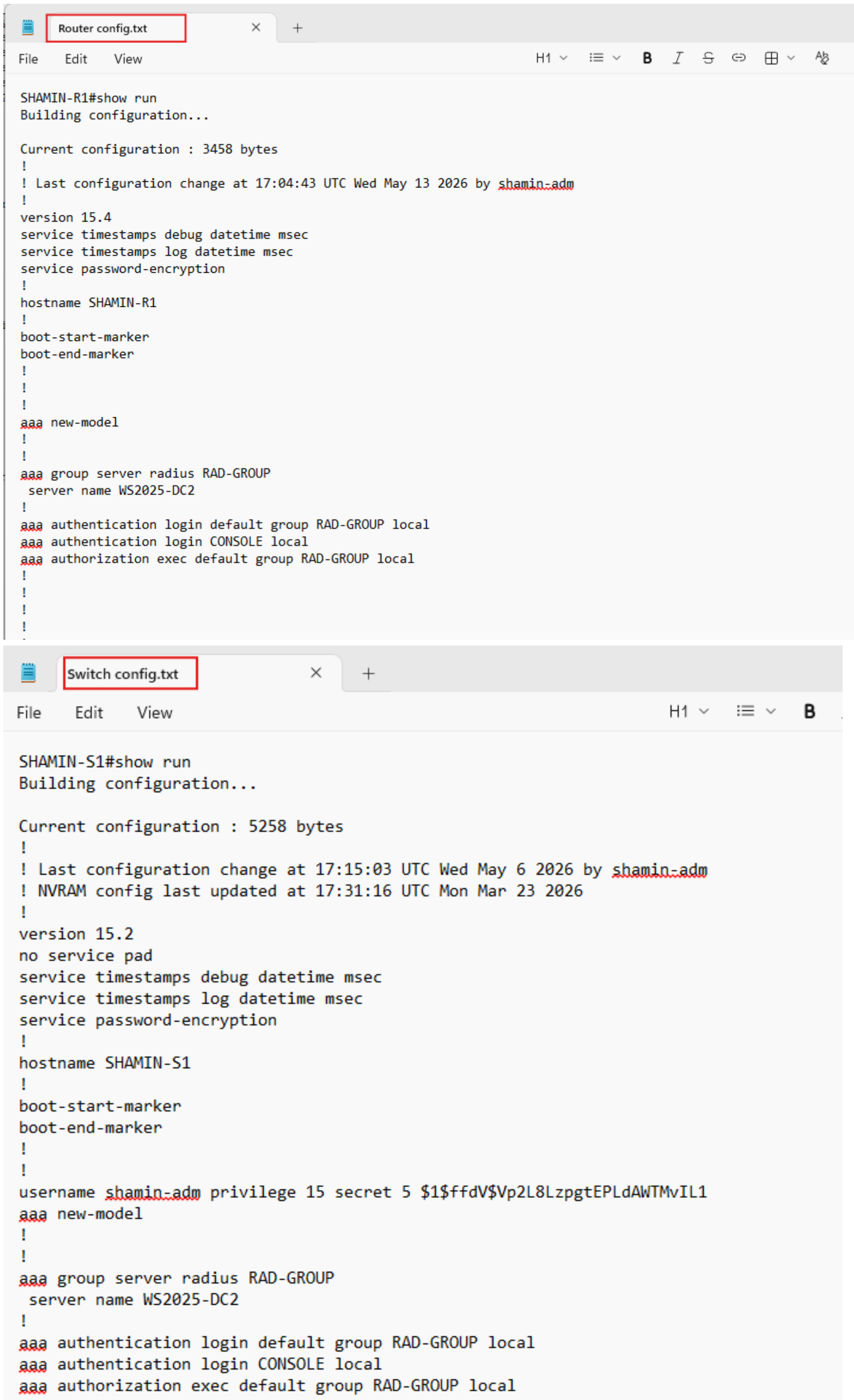
```
!
interface GigabitEthernet0/1
 ip address 172.16.1.115 255.255.255.0
 ip access-group DMZ-INBOUND in
 duplex auto
 speed auto
!
```

Figure 43 – Applying ACL rule on Routers G0/1 interface

6.5 Configuration Backup

To ensure reliability and recovery, configurations of the physical router and switch were backed up after completing all configurations.

This step is important in real-world environments to prevent data loss and allow quick restoration in case of failure.


Router config.txt

```
SHAMIN-R1#show run
Building configuration...

Current configuration : 3458 bytes
!
! Last configuration change at 17:04:43 UTC Wed May 13 2026 by shamin-adm
!
version 15.4
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname SHAMIN-R1
!
boot-start-marker
boot-end-marker
!
!
!
aaa new-model
!
!
aaa group server radius RAD-GROUP
 server name WS2025-DC2
!
aaa authentication login default group RAD-GROUP local
aaa authentication login CONSOLE local
aaa authorization exec default group RAD-GROUP local
!
!
!
!
```

Switch config.txt

```
SHAMIN-S1#show run
Building configuration...

Current configuration : 5258 bytes
!
! Last configuration change at 17:15:03 UTC Wed May 6 2026 by shamin-adm
! NVRAM config last updated at 17:31:16 UTC Mon Mar 23 2026
!
version 15.2
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname SHAMIN-S1
!
boot-start-marker
boot-end-marker
!
!
username shamin-adm privilege 15 secret 5 $1$ffdV$Vp2L8LzpgtEPLdAWTMvIL1
aaa new-model
!
!
aaa group server radius RAD-GROUP
 server name WS2025-DC2
!
aaa authentication login default group RAD-GROUP local
aaa authentication login CONSOLE local
aaa authorization exec default group RAD-GROUP local
!
```

Figure 44 – Backup configuration text file for Router and Switch

Part 7 – Additional Infrastructure and Security Services

7.1 Overview

Part 7 focused on expanding the virtual environment by implementing additional enterprise services and security tools. The completed tasks included Docker, Nagios, OpenVAS, WordPress, and Wazuh deployments within the ESXi infrastructure.

Initially, separate VLANs were created for each service environment. While this configuration functioned correctly, it introduced unnecessary complexity and routing overhead. A more efficient design would have been to place these internal service virtual machines under Virtual Server VLAN 30. However, since the environment was already fully configured and operational, the existing VLAN structure was kept for the remainder of the project.

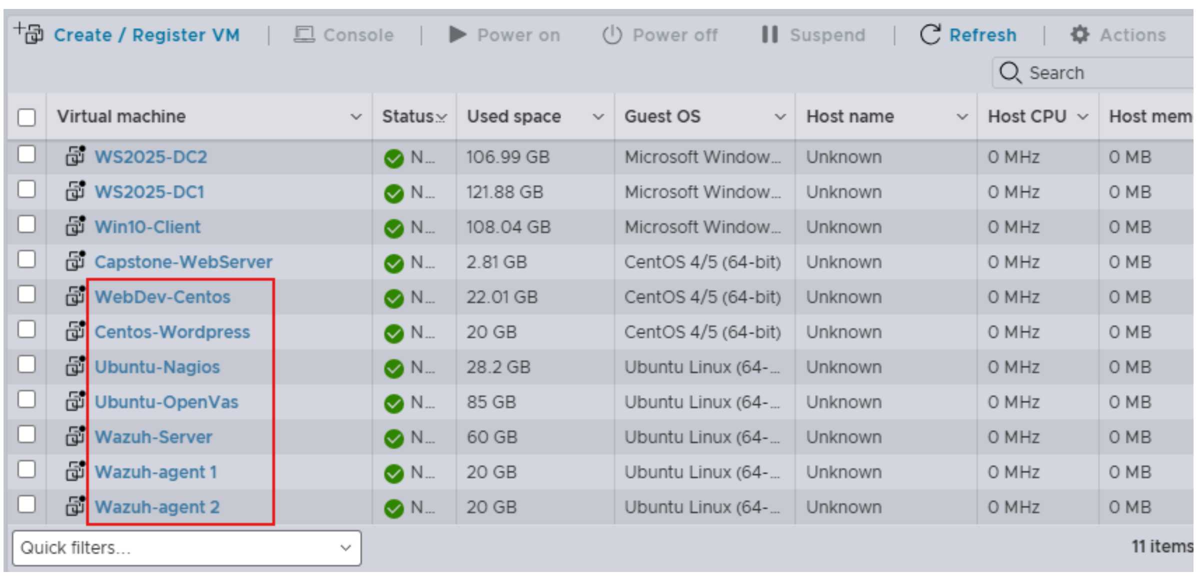


Figure 44 – Adding more services in the virtual environment.

7.2 Task 1 – Docker Web Development Environment

Docker was installed on a Centos virtual machine and configured to host a preconfigured Nginx web server container running on port 80. A dedicated development VLAN **(VLAN 60)** was used for this environment to isolate development traffic from the rest of the network.

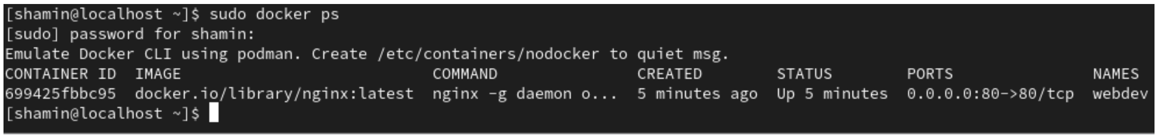
```
[shamin@localhost ~]$ ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group def
    link/ether 00:0c:29:e9:9f:62 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 10.115.60.10/24 brd 10.115.60.255 scope global noprefixroute ens33
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fee9:9f62/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: docker0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group de
    link/ether c6:f4:b3:f1:fb:9c brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
    inet6 fe80::c4f4:b3ff:fef1:fb9c/64 scope link
       valid_lft forever preferred_lft forever
```

Set static IP on this Centos VM

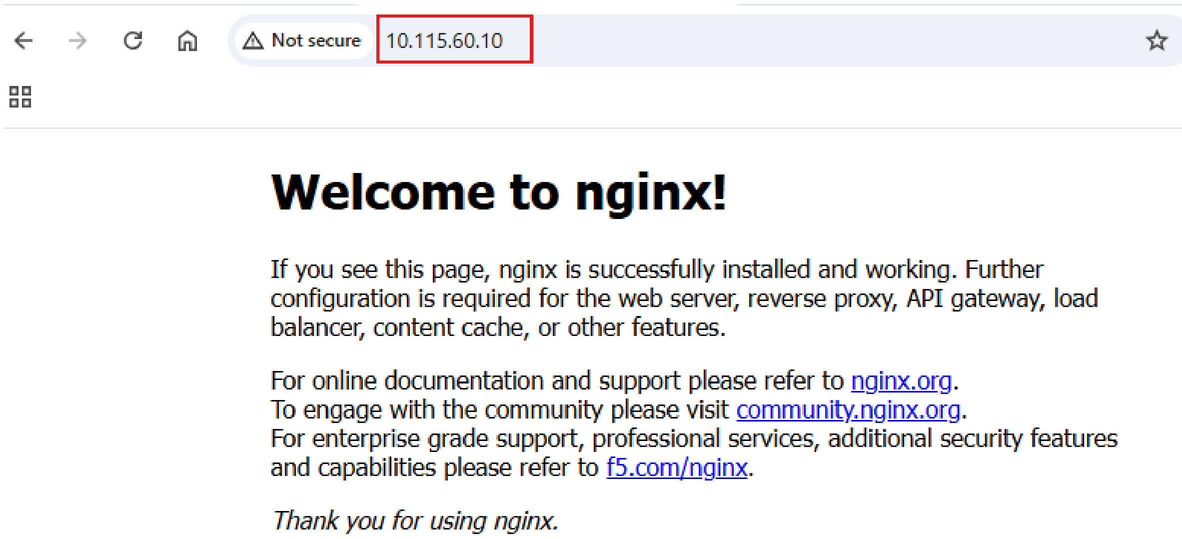
```
[shamin@localhost ~]$ sudo yum install docker -y

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for shamin:
CentOS Stream 9 - BaseOS
CentOS Stream 9 - AppStream
```

Successfully installing docker

```
[shamin@localhost ~]$ sudo docker run -d -p 80:80 --name webdev nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
801a1ad15b4e: Pull complete
3531af2bc2a9: Pull complete
ce776bbcda0d: Pull complete
85c66128325a: Pull complete
4677c2a9a3d4: Pull complete
ff048f1f2159: Pull complete
677c63196868: Pull complete
54c38c75806e: Download complete
69989ccd189b: Download complete
Digest: sha256:6e23479198b998e5e25921dff8455837c7636a67111a04a635cf1bb363d199dc
Status: Downloaded newer image for nginx:latest
0f9d351ce03575035ece17e7de305c8e64ca495e6a8898898155bb294e87341e
```



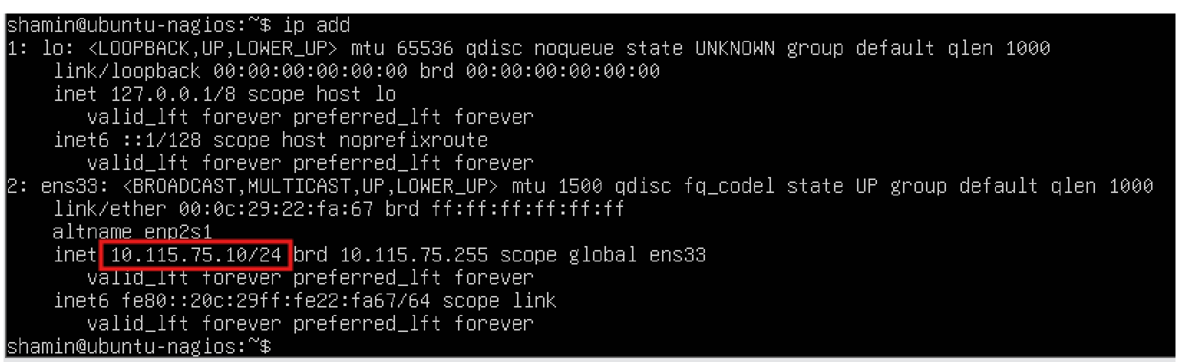
Official Nginx image created and running successfully inside the docker container



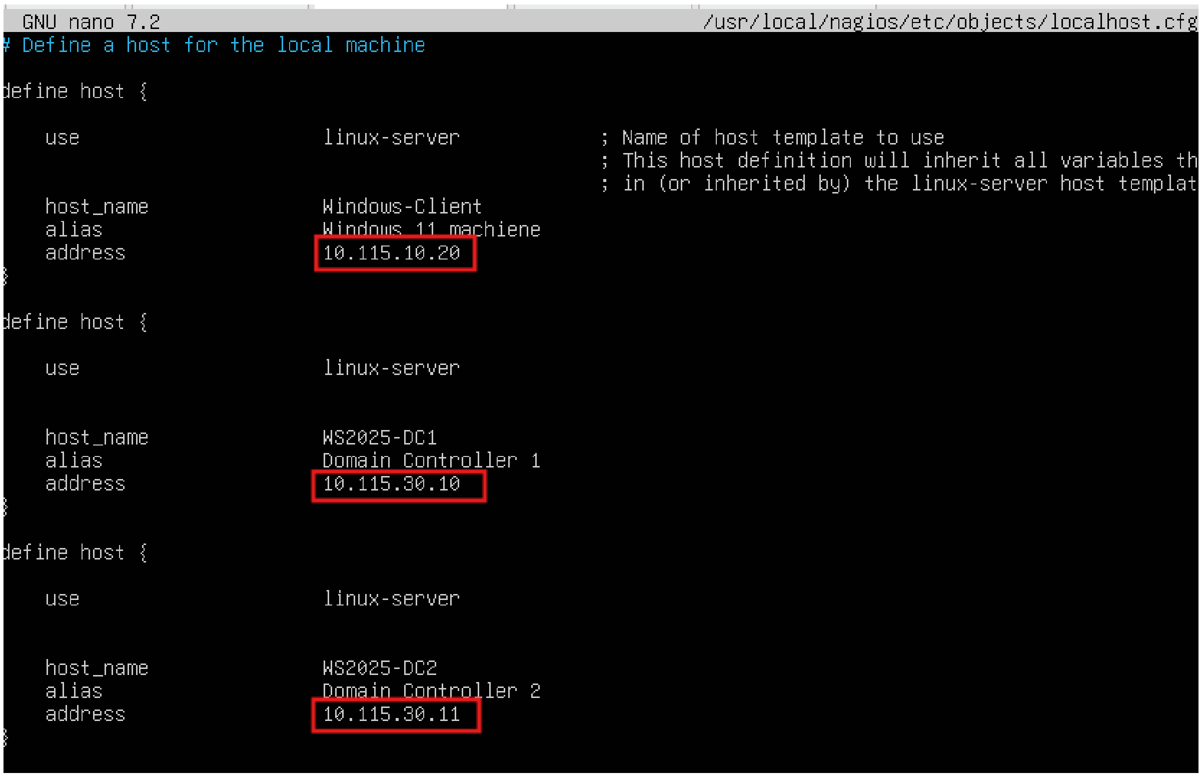
Internal access to Nginx web page

7.3 Task 2 – Nagios Monitoring Server

Nagios was installed on an Ubuntu virtual machine to monitor systems and services within the environment. Monitoring checks were configured for multiple devices, including server availability and network services.

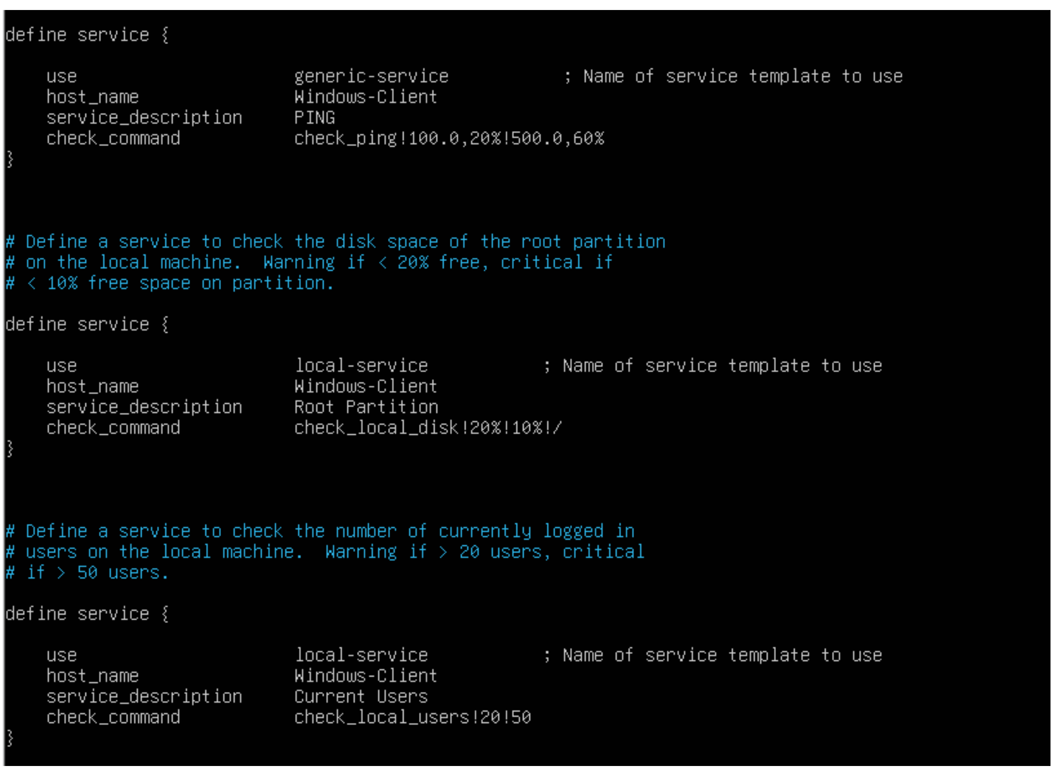


Set static IP on this ubuntu VM



```
GNU nano 7.2                                   /usr/local/nagios/etc/objects/localhost.cfg
# Define a host for the local machine

define host {

    use                     linux-server            ; Name of host template to use
                                                    ; This host definition will inherit all variables th
                                                    ; in (or inherited by) the linux-server host templat

    host_name               Windows-Client
    alias                   Windows 11 machiene
    address                 10.115.10.20
}

define host {

    use                     linux-server


    host_name               WS2025-DC1
    alias                   Domain Controller 1
    address                 10.115.30.10
}

define host {

    use                     linux-server


    host_name               WS2025-DC2
    alias                   Domain Controller 2
    address                 10.115.30.11
}
```

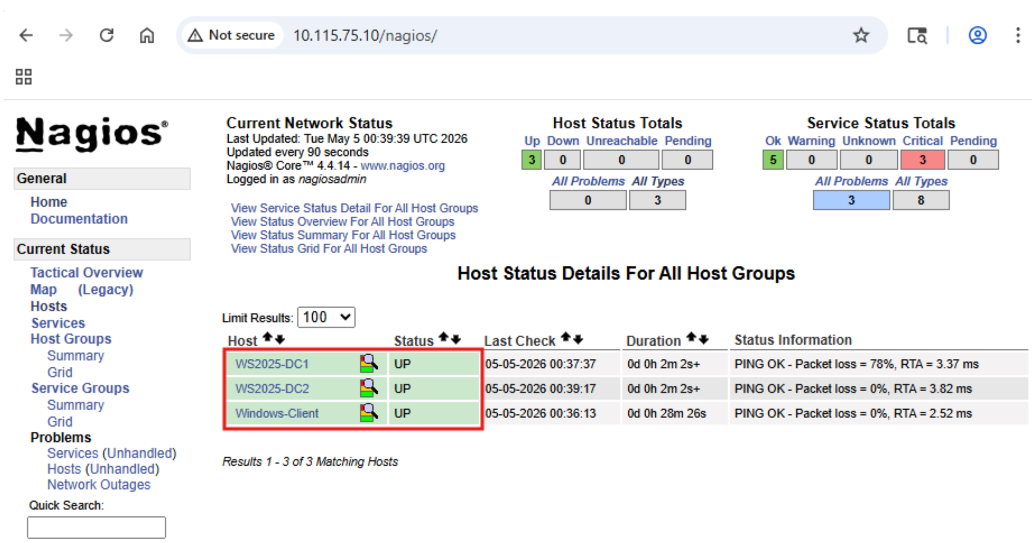
Editing Nagios configuration file to add host machines for monitoring



```
define service {

    use                     generic-service           ; Name of service template to use
    host_name               Windows-Client
    service_description     PING
    check_command           check_ping!100.0,20%!500.0,60%
}


# Define a service to check the disk space of the root partition
# on the local machine.  Warning if < 20% free, critical if
# < 10% free space on partition.

define service {

    use                     local-service           ; Name of service template to use
    host_name               Windows-Client
    service_description     Root Partition
    check_command           check_local_disk!20%!10%!/
}


# Define a service to check the number of currently logged in
# users on the local machine.  Warning if > 20 users, critical
# if > 50 users.

define service {

    use                     local-service           ; Name of service template to use
    host_name               Windows-Client
    service_description     Current Users
    check_command           check_local_users!20!50
}
```

Defining specific monitoring services in Nagios

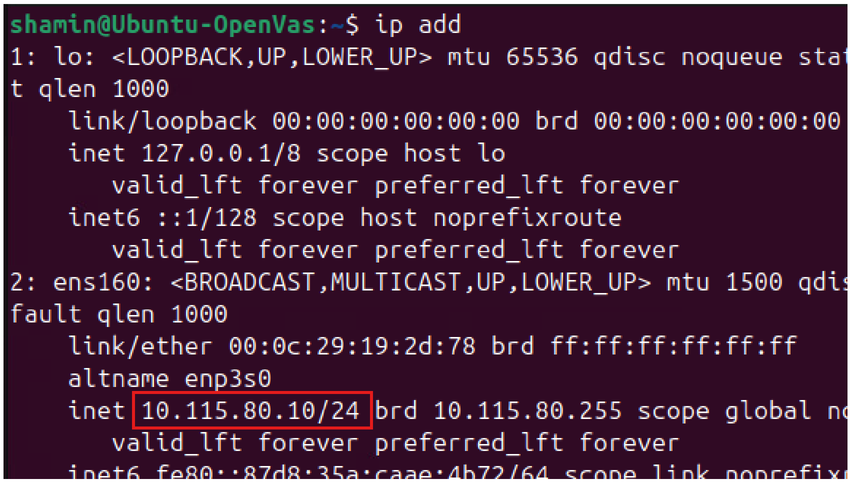
Successfully access Nagios dashboard



Nagios Core web interface dashboard displaying the monitoring status of my network hosts.

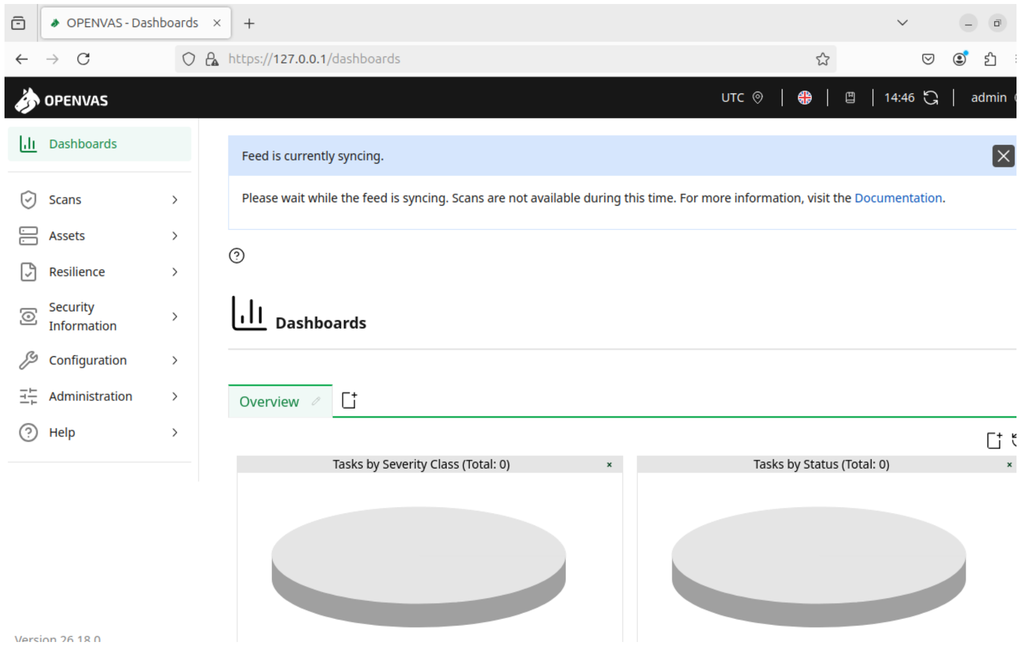
7.4 Task 3 – OpenVAS Vulnerability Scanner

OpenVAS was installed using docker on an Ubuntu virtual machine and configured to perform vulnerability scans against systems within the environment.

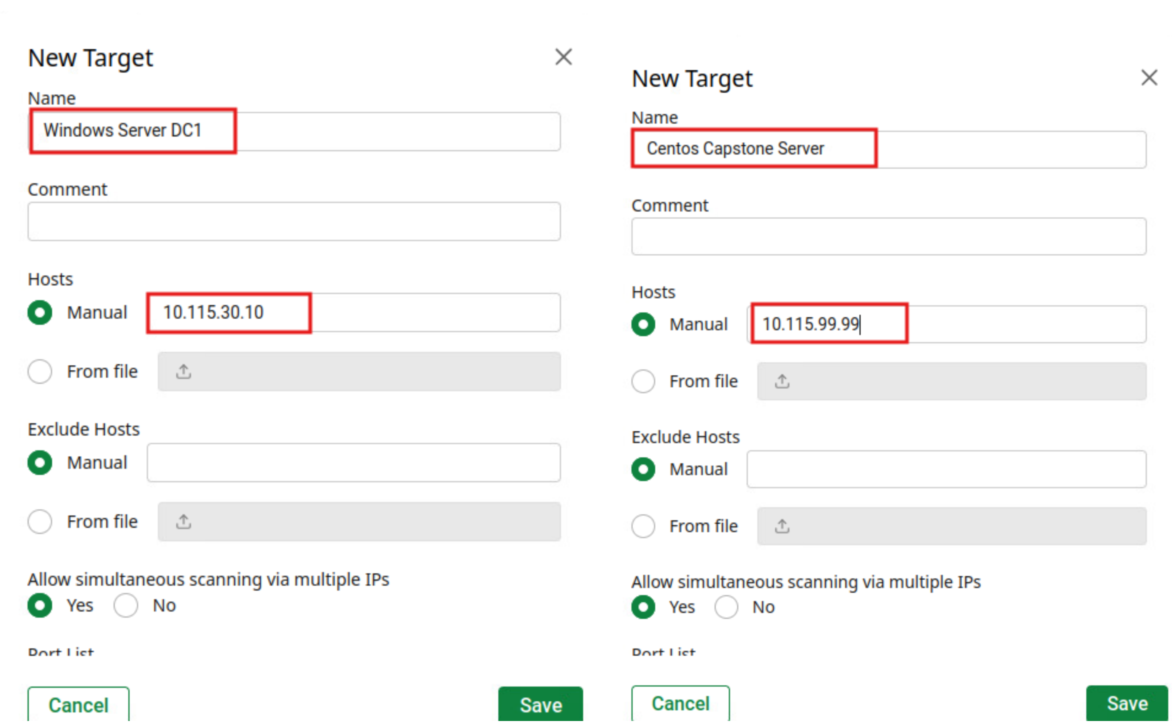


Setting static IP on this Ubuntu VM

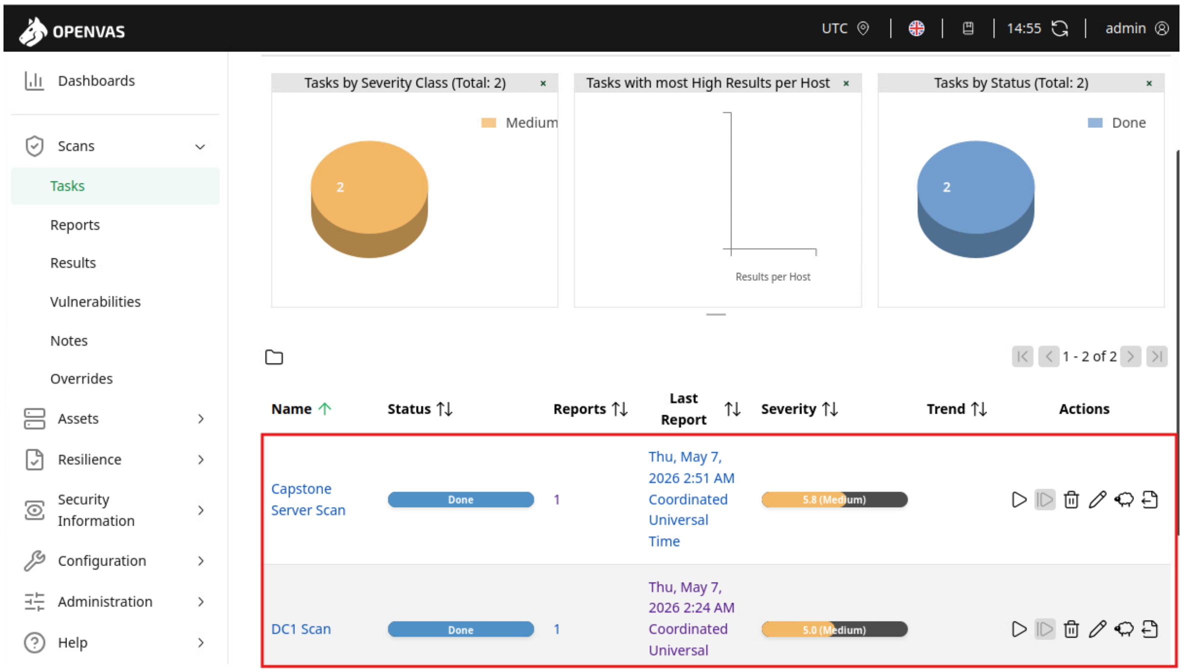
It shows a successful deployment of the Greenbone Community Edition **(OpenVAS)** using Docker Compose



Access OpenVas dashboard



Making Windows server and DMZ server the target for vulnerability scan



Vulnerability scan results for Windows server and DMZ server, both had medium severity rate

7.5 Task 4 – WordPress Web Server

WordPress was deployed on a Centos virtual machine and configured with a custom website template. The website was made accessible to other students through the network environment.

```
[shamin@localhost ~]$ ip add
: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP grou
    link/ether 00:0c:29:d5:53:47 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 10.115.70.10/24 brd 10.115.70.255 scope global noprefixroute ens33
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fed5:5347/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[shamin@localhost ~]$
```

Set static IP

```
[shamin@localhost ~]$ sudo dnf install httpd mariadb-server mariadb -y

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for shamin:
CentOS Stream 9 - BaseOS                        4.6 MB/s | 8.9 MB     00:01
CentOS Stream 9 - AppStream
CentOS Stream 9 - Extras packages
Dependencies resolved.
================================================================================
 Package                                        Architecture
================================================================================
Installing:
 httpd                                          x86_64
 mariadb                                        x86_64
 mariadb-server                                 x86_64
```

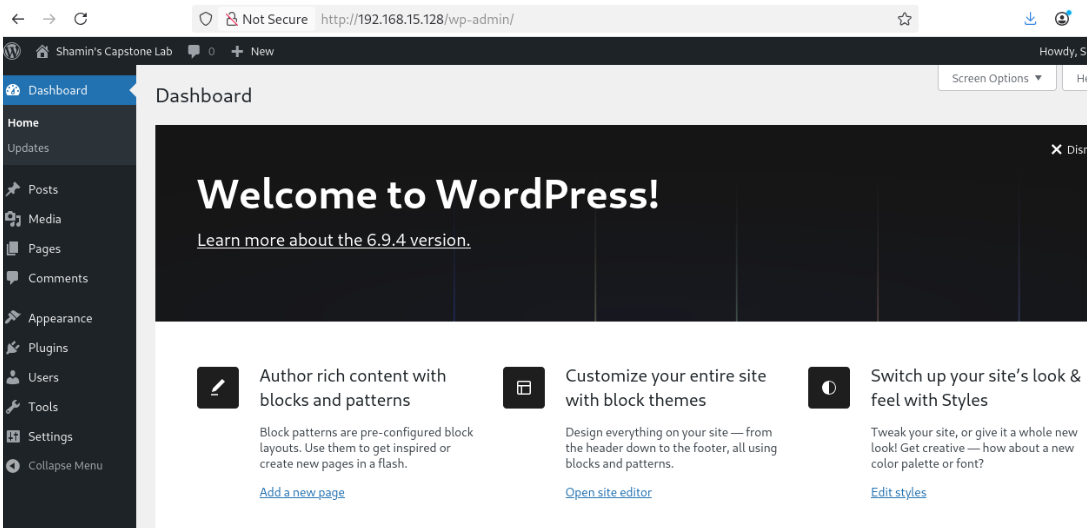
Installing LAMP stack (Linux, Apache, MySQL, PHP)

```
[shamin@localhost ~]$ sudo dnf install php php-mysqlnd php-gd php-xml php-mbstring -y
Last metadata expiration check: 0:00:47 ago on Wed 29 Apr 2026 01:03:30 PM EDT.
Dependencies resolved.
==========================================================================================
 Package                                  Architecture                    Version
==========================================================================================
Installing:
 php                                      x86_64                          8.0.30-5.el9
 php-gd                                   x86_64                          8.0.30-5.el9
 php-mbstring                             x86_64                          8.0.30-5.el9
 php-mysqlnd                              x86_64                          8.0.30-5.el9
 php-xml                                  x86_64                          8.0.30-5.el9
```

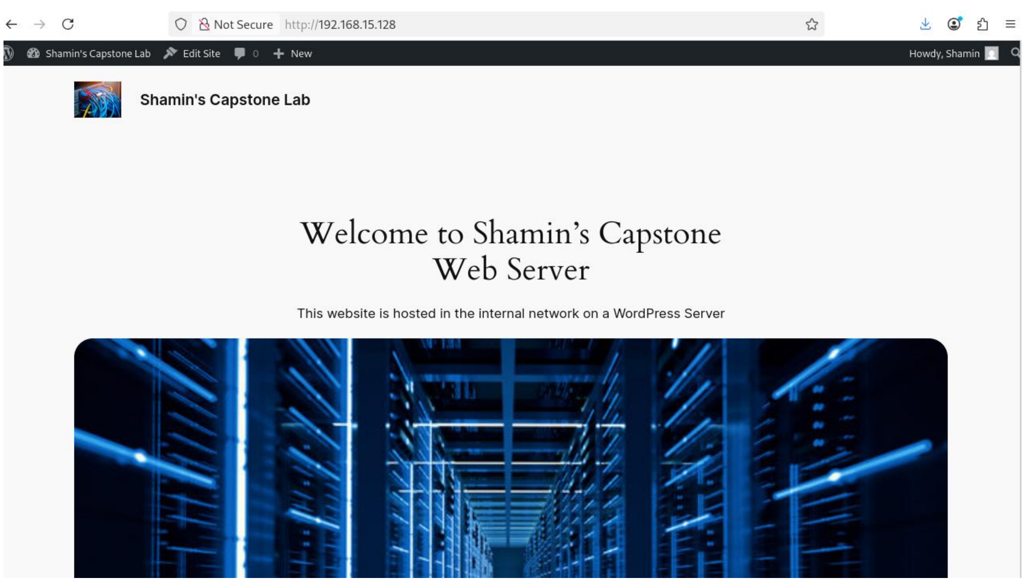
Installing PHP to run WordPress on my CentOS stream 9

```
[shamin@localhost ~]$ wget https://wordpress.org/latest.tar.gz
--2026-04-29 13:21:50--  https://wordpress.org/latest.tar.gz
Resolving wordpress.org (wordpress.org)... 198.143.164.252, 2607:f978:5:8002::c68f:a4fc
Connecting to wordpress.org (wordpress.org)|198.143.164.252|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 27064862 (26M) [application/octet-stream]
Saving to: 'latest.tar.gz'

latest.tar.gz                    100%[============================================

2026-04-29 13:21:52 (32.4 MB/s) - 'latest.tar.gz' saved [27064862/27064862]

[shamin@localhost ~]$ tar -xvzf latest.tar.gz
wordpress/
wordpress/index.php
wordpress/license.txt
wordpress/readme.html
wordpress/wp-activate.php
wordpress/wp-admin/
```

Downloading the official WordPress application files and extracting them on my server



Accessing WordPress administration dashboard



Created this website using WordPress template and accessing it from classroom computer

```
Extended IP access list DMZ-INBOUND
    10 permit ip 172.16.2.0 0.0.0.255 any (9784 matches)
    20 permit tcp any host 10.115.99.99 eq www (371 matches)
    25 permit tcp any host 10.115.70.10 eq www (105 matches)
    30 permit tcp any host 10.115.99.99 eq 443 (180 matches)
    35 permit tcp any host 10.115.70.10 eq 443
    40 deny ip any 10.115.0.0 0.0.255.255
    50 permit ip any any (1851358 matches)
SHAMIN-R1#
```

Configuring the existing ACL rules on router to make this website accessible to other students

7.6 Task 6 – Wazuh Security Monitoring

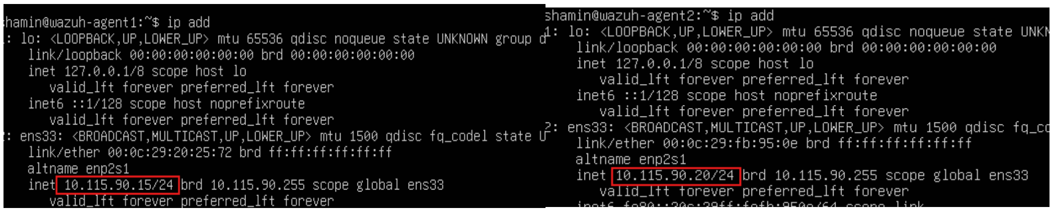
Wazuh was installed on an Ubuntu server using docker to provide centralized security monitoring and log analysis. Two additional Ubuntu virtual machines were configured as Wazuh agents to forward logs and security events to the Wazuh server.

```
       valid_lft forever preferred_lft forever
shamin@ubuntu-wazuh:~$ ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qle
    link/ether 00:0c:29:e4:a7:26 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 10.115.90.10/24 brd 10.115.90.255 scope global ens33
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fee4:a726/64 scope link
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group defaul
    link/ether 26:08:50:d6:fa:d2 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
```

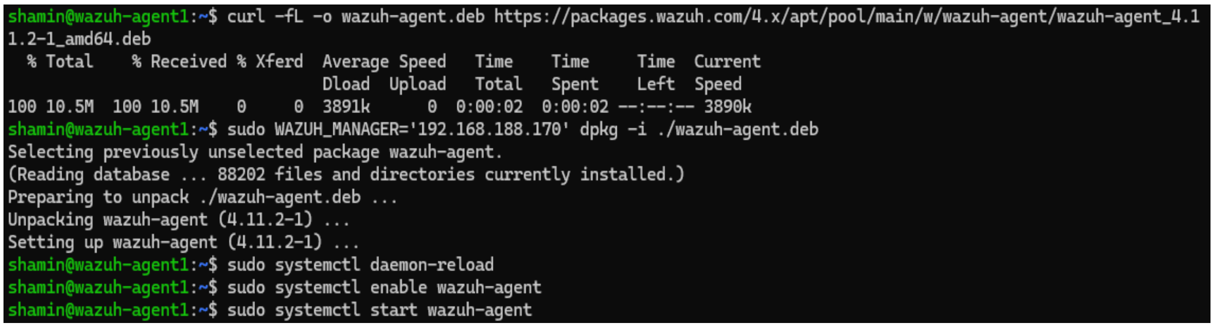
Set a static IP on Ubuntu-Wazuh server

```
shamin@ubuntu-wazuh:~$ sudo docker ps
CONTAINER ID   IMAGE                              COMMAND                  CREATED          STATUS

5d18b417ef6e   wazuh/wazuh-dashboard:4.11.2       "/entrypoint.sh"         11 hours ago     Up 11 minute

df5089ef4834   wazuh/wazuh-manager:4.11.2         "/init"                  11 hours ago     Up 11 minute
5/tcp, 0.0.0.0:514->514/udp, [::]:514->514/udp, 0.0.0.0:55000->55000/tcp, [::]:55000->55000/tcp, 1
6afd51121e2e   wazuh/wazuh-indexer:4.11.2         "/entrypoint.sh open…"   11 hours ago     Up 11 minute

shamin@ubuntu-wazuh:~$
```

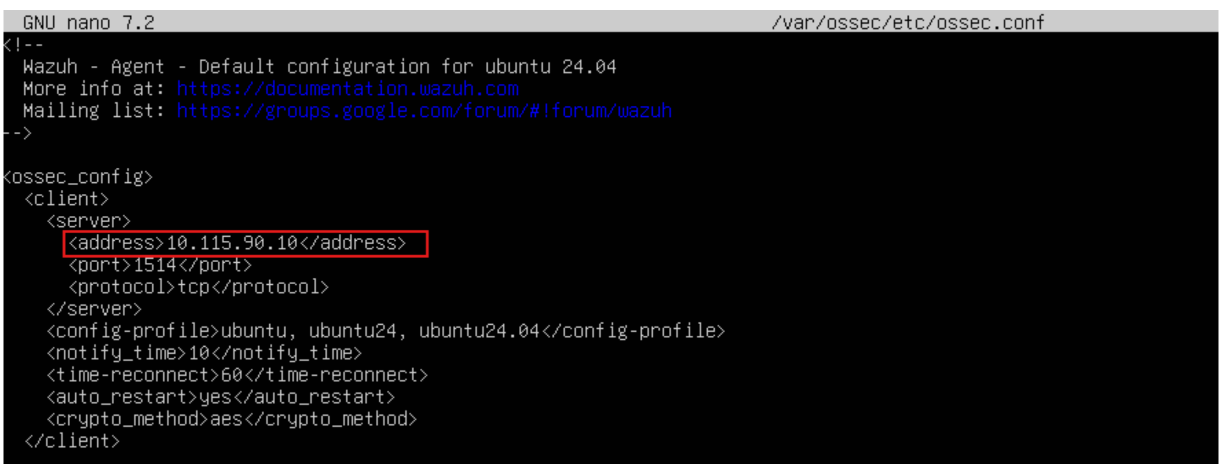
It shows the successful deployment of a single-node Wazuh SIEM stack running inside Docker containers



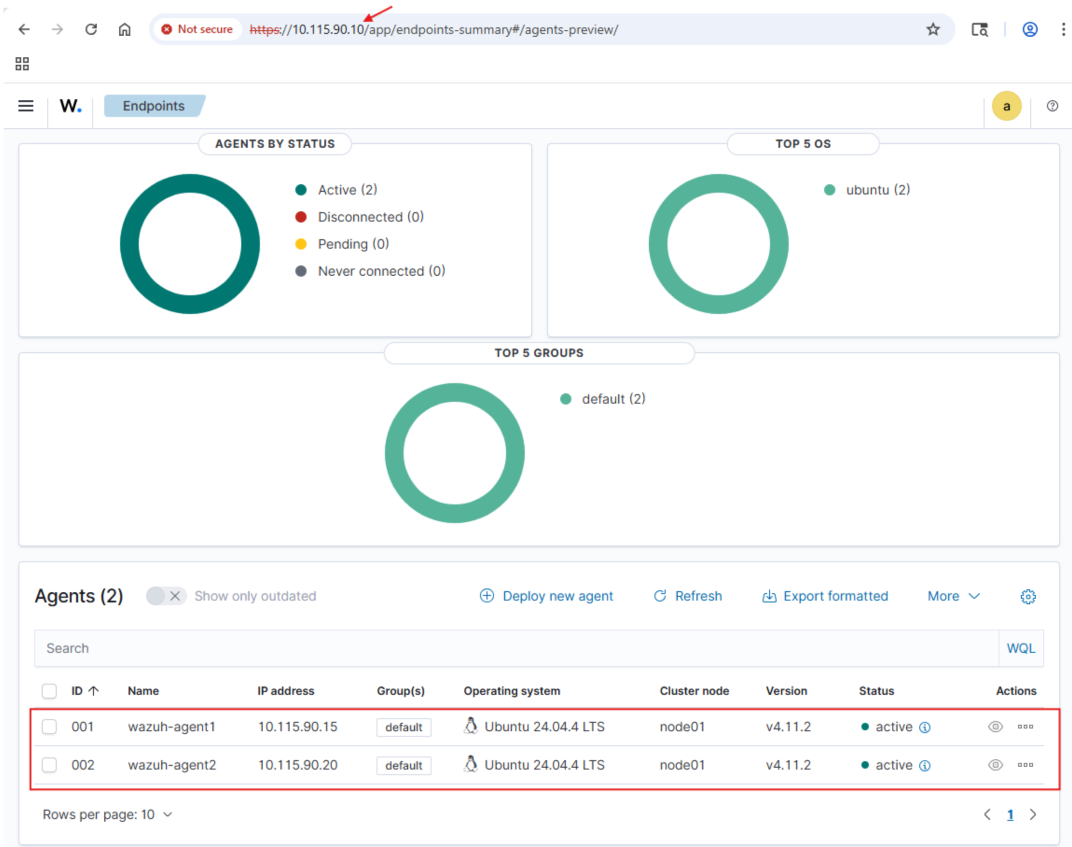
Set Static IP for both wazuh-agent1 and wazuh-agent2 VM



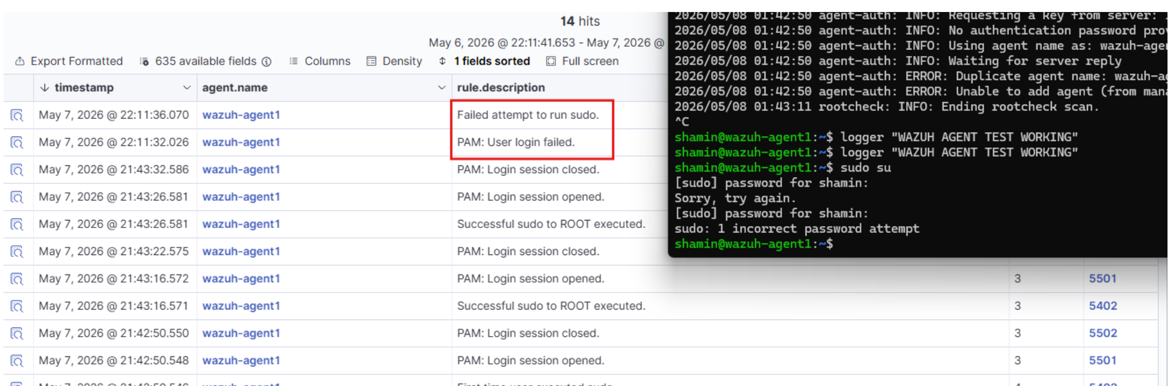
It shows successful downloading, installing, and starting the Wazuh Agent on wazuh-agent1 and also wazuh-agent2 VM.



Editing configuration file for the wazuh agent to set IP address of Wazuh manager, that this wazuh agent VMs will report to



This confirms that my Wazuh Manager is now successfully monitoring both Wazuh Agents VMs in my network



Some real-time security log monitoring and alert generation in action.